

Verifying Rust Code with Verus

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<https://github.com/verus-lang/verus/>

Contributors 84



[+ 70 contributors](#)

Verus: Floyd-Hoare Logic for Rust

Verus is:

- ... an implementation of **Floyd-Hoare logic**
- ... for the **Rust** programming Language
- ... using **SMT solvers** to automate proofs
- ... adding some higher-order logic, extensional =, induction, etc.

Inspired heavily by Dafny and F*

Similar Rust verification tools: Kani, Creusot, Prusti, Aeneas, ...

Verus demo

```
C:\verus\source> verus.exe binsearch.rs  
verification results:: 2 verified, 0 errors
```

```
fn binary_search(v: &Vec<u64>, key: u64) -> (ret: usize)  
  requires  
    forall |i: int, j: int| 0 <= i <= j < v.len() ==> v[i] <= v[j],  
    exists |i: int| 0 <= i < v.len() && key == v[i],  
  ensures  
    ret < v.len(),  
    key == v[ret as int],  
{  
  let mut left: usize = 0;  
  let mut right: usize = v.len() - 1;  
  while left != right  
    invariant  
      right < v.len(),  
      exists |i: int| left <= i <= right && key == v[i],  
      forall |i: int, j: int| 0 <= i <= j < v.len() ==> v[i] <= v[j],  
    decreases right - left  
  {  
    let mid = left + (right - left) / 2;  
    if v[mid] < key {  
      left = mid + 1;  
    } else {  
      right = mid;  
    }  
  }  
  left  
}
```

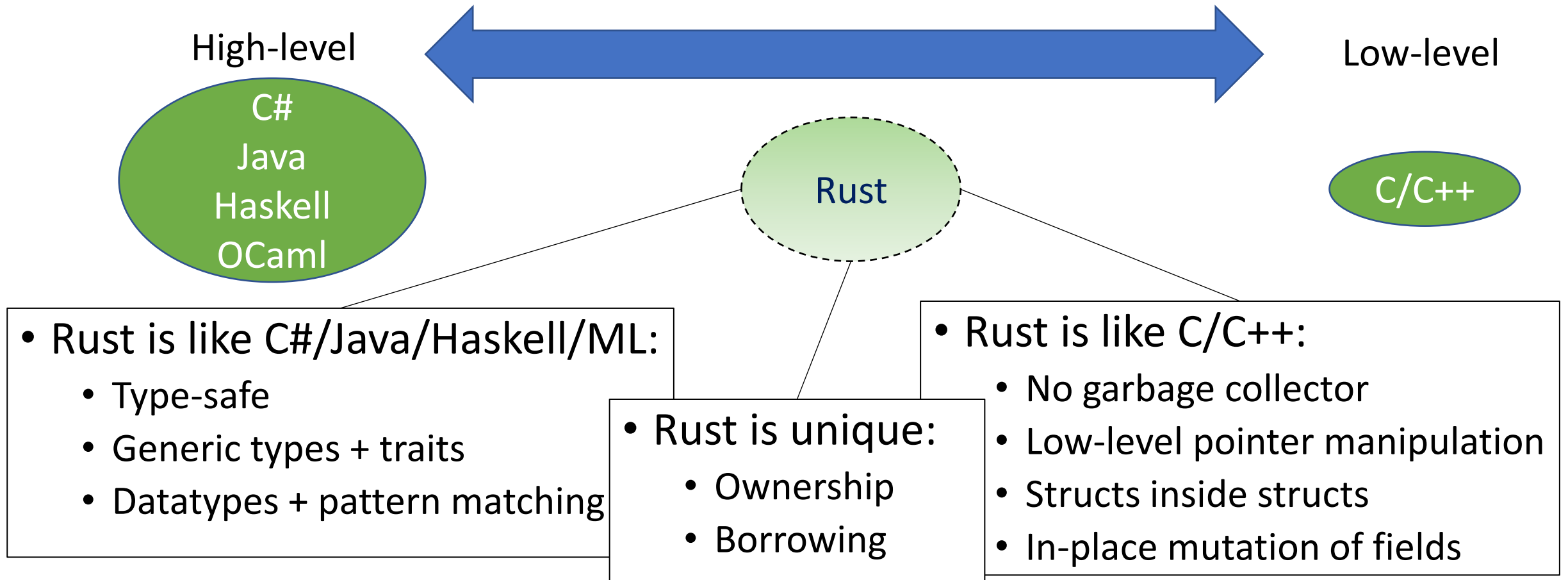
Verus demo

```
C:\verus\source> verus.exe binsearch.rs
error: invariant not satisfied at end of loop body
--> .\scratch.rs:29:13
29 |     exists|i: int| left <= i <= right && key == v[i],
    |     ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
error: aborting due to previous error

verification results:: 1 verified, 1 errors
```

```
fn binary_search(v: &Vec<u64>, key: u64) -> (ret: usize)
  requires
    forall|i: int, j: int| 0 <= i <= j < v.len() ==> v[i] <= v[j],
    exists|i: int| 0 <= i < v.len() && key == v[i],
  ensures
    ret < v.len(),
    key == v[ret as int],
{
  let mut left: usize = 0;
  let mut right: usize = v.len() - 1;
  while left != right
    invariant
      right < v.len(),
      exists|i: int| left <= i <= right && key == v[i],
      forall|i: int, j: int| 0 <= i <= j < v.len() ==> v[i] <= v[j],
      decreases right - left
    {
      let mid = left + (right - left) / 2;
      if v[mid] < key {
        left = mid + 2;
      } else {
        right = mid;
      }
    }
  left
}
```

Why write systems code in Rust?



Why write systems code in Rust?

“Speaking of languages, it's time to halt starting any new projects in C/C++ and use Rust for those scenarios where a non-GC language is required. For the sake of security and reliability, the industry should declare those languages as deprecated.”

- Mark Russinovich (Microsoft Azure CTO), 2022

Goal: verification for systems programmers

- No separate proof language or proof assistant
- specs + proofs + executable code in Rust

```
spec fn triangle(n: int) -> int
  decreases n
{
  if n <= 0 {
    0
  } else {
    n + triangle(n - 1)
  }
}
```

```
proof fn triangle_is_monotonic(i: int, j: int)
  ensures
    i <= j ==> triangle(i) <= triangle(j),
  decreases j
{
  if j > 0 {
    triangle_is_monotonic(i, j - 1);
  }
}
```

```
fn loop_triangle(n: u32) -> (sum: u32)
  requires triangle(n as int) <= u32::MAX
  ensures sum == triangle(n as int)
{
  let mut sum: u32 = 0;
  for idx in 0..n
    invariant
      sum == triangle(idx as int),
      triangle(n as int) <= u32::MAX,
    {
      assert(sum + (idx + 1) <= u32::MAX) by {
        triangle_is_monotonic(idx + 1, n as int);
      }
      sum = sum + (idx + 1);
    }
  sum
}
```

Note: spec fn, proof fn must terminate

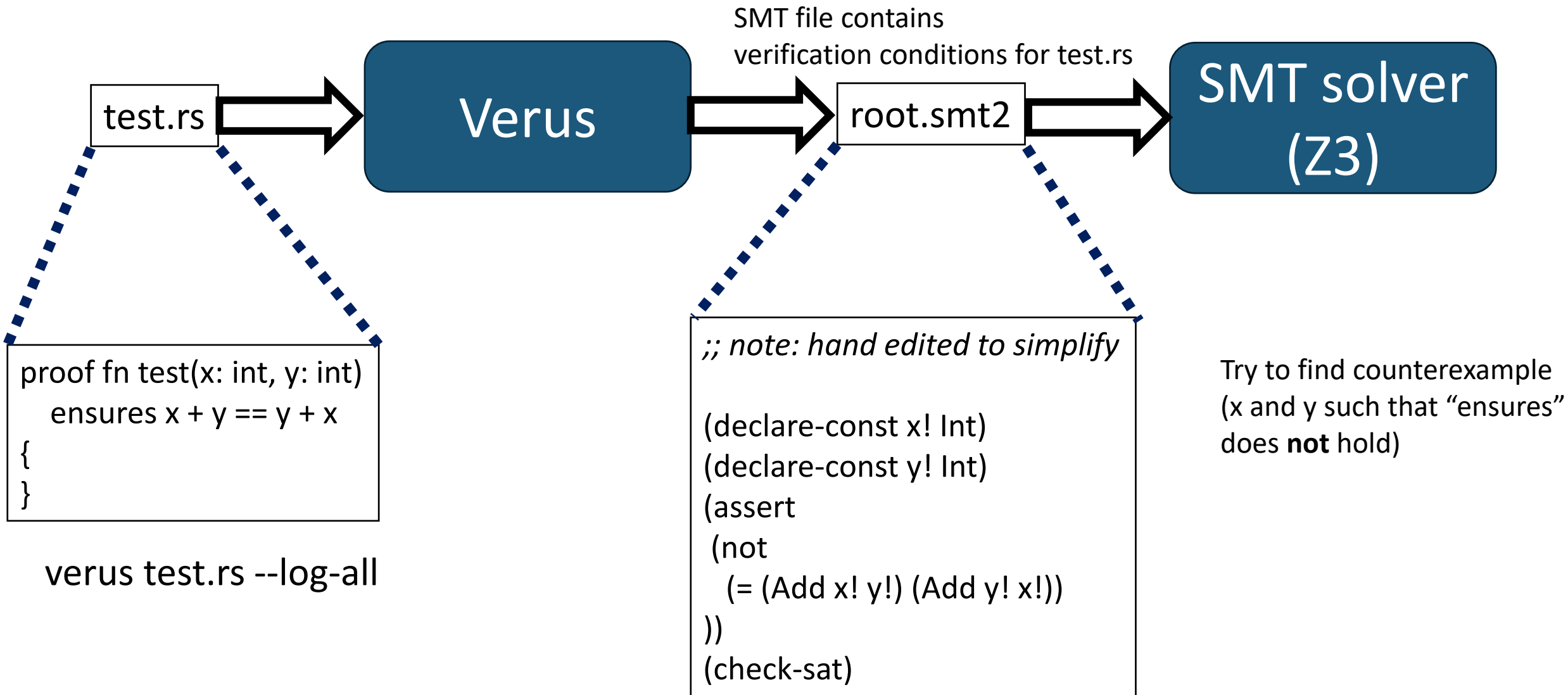
- disallowed: spec fn f() -> bool { !f() }
- disallowed: proof fn f() ensures false { f(); }

Outline of lectures

- Verification condition (VC) generation
- Using SMT solvers
- Higher-order features, vstd library
- Generics and traits (briefly)
- Ownership, &, &mut
- Verifying unsafe Rust code
- Concurrency

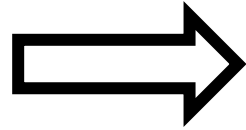
Verification condition (VC) generation

Verification condition generation

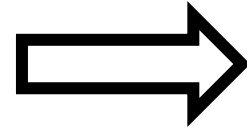


Verification condition generation

test.rs



root.air



root.smt2

```
proof fn test1(i: int) -> (r: int)
  ensures r == i + 15
{
  let mut x = i + 3;
  x = x + 5;
  x = x + 7;
  x
}
```

```
;; note: hand edited to simplify
(check-valid
  (declare-const r! Int)
  (declare-const i! Int)
  (declare-var x@ Int)

  (block
    (assume (= x@ (Add i! 3)))
    (assign x@ (Add x@ 5))
    (assign x@ (Add x@ 7))
    (assume (= r! x@))
    (assert (= r! (Add i! 15)))
  )
)
```

```
;; note: hand edited to simplify
(declare-const r! Int)
(declare-const i! Int)
(declare-const x@0 Int)
(declare-const x@1 Int)
(declare-const x@2 Int)
(assert
  (not
    (=> (= x@0 (Add i! 3))
      (=> (= x@1 (Add x@0 5))
        (=> (= x@2 (Add x@1 7))
          (=> (= r! x@2)
            (= r! (Add i! 15))
          )
        )
      )
    )
  )
)
(check-sat)
```

Weakest preconditions

From Speaking Logic lecture:

The proof of a valid triple $\{P\}S\{Q\}$ can be decomposed into

1. The valid triple $\{wlp(S)(Q)\}S\{Q\}$, and
2. The valid assertion $P \implies wlp(S)(Q)$

```
wlp({
  assume x = i + 3;
  assign x := x + 5;
  assign x := x + 7;
  assume r = x;
  assert r = i + 15
},
true)
```

```
wlp((assert E), Q) = E && Q
wlp((assume E), Q) = E ==> Q
wlp((assign x := E), Q) = subst x := E in Q
wlp((S1; S2), Q) = wlp(S1, wlp(S2, Q))
```

```
wlp(assume x = i + 3,
wlp(assign x := x + 5,
wlp(assign x := x + 7,
wlp(assume r = x,
wlp(assert r = i + 15,
true))))))
```

```
wlp(assume x = i + 3,
wlp(assign x := x + 5,
wlp(assign x := x + 7,
wlp(assume r = x,
  r = i + 15 && true
))))
```

```
wlp(assume x = i + 3,
wlp(assign x := x + 5,
wlp(assign x := x + 7,
  r = x ==>
  r = i + 15 && true
)))
```

```
wlp(assume x = i + 3,
wlp(assign x := x + 5,
  r = x + 7 ==>
  r = i + 15 && true
))
```

```
wlp(assume x = i + 3,
  r = (x + 5) + 7 ==>
  r = i + 15 && true
)
```

```
x = i + 3 ==>
r = (x + 5) + 7 ==>
r = i + 15 && true
```

Weakest preconditions via SSA (single static assignment)

```
wlp({
  assume  $x_0 = i + 3$ ;
  assign  $x_1 := x_0 + 5$ ;
  assign  $x_2 := x_1 + 7$ ;
  assume  $r = x_2$ ;
  assert  $r = i + 15$ 
},
true)
```

$wlp((\text{assert } E), Q) = E \ \&\& \ Q$
 $wlp((\text{assume } E), Q) = E \implies Q$
 $wlp((\text{assign } x_n := E), Q) = x_n = E \implies Q$
 $wlp((S1; S2), Q) = wlp(S1, wlp(S2, Q))$

```
 $x_0 = i + 3 \implies$ 
 $x_1 = x_0 + 5 \implies$ 
 $x_2 = x_1 + 7 \implies$ 
 $r = x_2 \implies$ 
 $r = i + 15 \ \&\& \ \text{true}$ 
```

```
wlp(assume  $x_0 = i + 3$ ,
wlp(assign  $x_1 := x_0 + 5$ ,
wlp(assign  $x_2 := x_1 + 7$ ,
wlp(assume  $r = x_2$ ,
wlp(assert  $r = i + 15$ ,
true))))))
```

```
wlp(assume  $x_0 = i + 3$ ,
wlp(assign  $x_1 := x_0 + 5$ ,
wlp(assign  $x_2 := x_1 + 7$ ,
wlp(assume  $r = x_2$ ,
 $r = i + 15 \ \&\& \ \text{true}$ 
))))
```

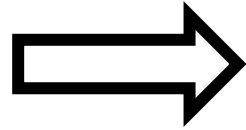
```
wlp(assume  $x_0 = i + 3$ ,
wlp(assign  $x_1 := x_0 + 5$ ,
wlp(assign  $x_2 := x_1 + 7$ ,
 $r = x_2 \implies$ 
 $r = i + 15 \ \&\& \ \text{true}$ 
)))
```

```
wlp(assume  $x_0 = i + 3$ ,
wlp(assign  $x_1 := x_0 + 5$ ,
 $x_2 = x_1 + 7 \implies$ 
 $r = x_2 \implies$ 
 $r = i + 15 \ \&\& \ \text{true}$ 
))
```

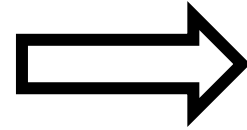
```
wlp(assume  $x_0 = i + 3$ ,
 $x_1 = x_0 + 5 \implies$ 
 $x_2 = x_1 + 7 \implies$ 
 $r = x_2 \implies$ 
 $r = i + 15 \ \&\& \ \text{true}$ 
)
```

Verification condition generation

test.rs



root.air



root.smt2

```
proof fn test1(i: int) -> (r: int)
  ensures r == i + 15
{
  let mut x = i + 3;
  x = x + 5;
  x = x + 7;
  x
}
```

```
;; note: hand edited to simplify
(check-valid
  (declare-const r! Int)
  (declare-const i! Int)
  (declare-var x@ Int)

  (block
    (assume (= x@ (Add i! 3)))
    (assign x@ (Add x@ 5))
    (assign x@ (Add x@ 7))
    (assume (= r! x@))
    (assert (= r! (Add i! 15)))
  )
)
```

```
;; note: hand edited to simplify
(declare-const r! Int)
(declare-const i! Int)
(declare-const x@0 Int)
(declare-const x@1 Int)
(declare-const x@2 Int)
(assert
  (not
    (=> (= x@0 (Add i! 3))
      (=> (= x@1 (Add x@0 5))
        (=> (= x@2 (Add x@1 7))
          (=> (= r! x@2)
            (= r! (Add i! 15))
          )
        )
      )
    )
  )
)
(check-sat)
```

```
x0 = i + 3 ==>
x1 = x0 + 5 ==>
x2 = x1 + 7 ==>
r = x2 ==>
r = i + 15 && true
```

Using SMT solvers

Decidable and undecidable theories

DECIDABLE THEORIES

boolean expressions

$e ::= \text{true} \mid \text{false} \mid !e \mid e \ \&\& \ e \mid e \implies e \mid \dots$

linear integer arithmetic

$e ::= \dots \mid -2 \mid -1 \mid 0 \mid 1 \mid 2 \mid e + e \mid e - e \mid e \leq e \mid \dots$

bit vector arithmetic

$e ::= \dots \mid e \ \& \ e \mid e \ll e \mid \dots$

uninterpreted functions

$e ::= \dots \mid f(e, \dots, e)$

datatypes

$e ::= \dots \mid S(e, \dots, e) \mid S.f \mid \text{match } e \{ \dots \}$

UNDECIDABLE

nonlinear integer arithmetic

$e ::= \dots \mid e * e \mid e / e \mid e \% e$

quantifiers

$e ::= \dots \mid (\text{forall } x: t \mid e) \mid (\text{exists } x: t \mid e)$

What algorithms solve the decidable theories?

What heuristics can help with the undecidable components?

SMT components used by Verus

Core Components (always enabled)

SAT solver

true | false | !e
e && e | ...

Linear int arith

-2 | -1 | 0 | 1 | 2
| e + e | e - e
| e <= e | ...

Uninterp funcs

spec fn f(x,...,x);
f(e,...,e)

Datatypes

struct S { ... }
enum E { ... }
S(e,...,e) | e.f | ...

Quantifiers

forall | x: t | e
exists | x: t | e

Additional Components (via “assert by”)

Bit vector arith

e & e | e << e |
...

Nonlin int arith

e * e
e / e | e % e

decidable, but can be slow

undecidable,
often unpredictable

- undecidable, but really important
- programmable by triggers

Congruence for uninterpreted functions

Uninterp funcs

```
spec fn f(x,...,x);  
f(e,...,e)
```

Congruence principle:
equal inputs imply equal output

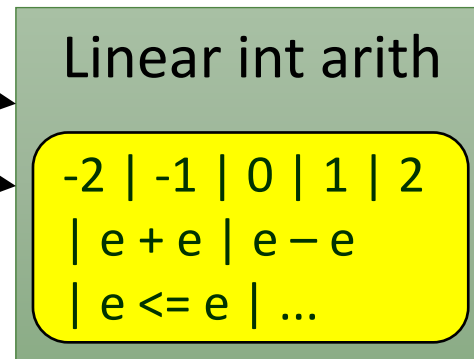
$$\left. \begin{array}{l} e1 == e1' \\ e2 == e2' \\ \dots \\ en == en' \end{array} \right\} \implies f(e1, \dots, en) == f(e1', \dots, en')$$

Example: uninterp funcs + arithmetic

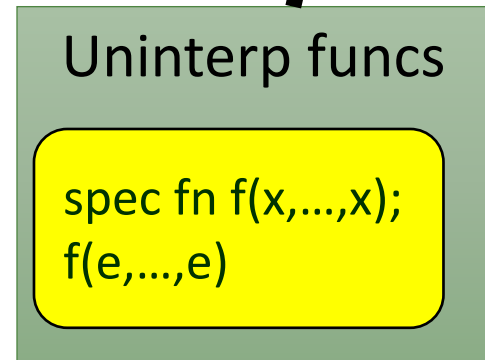
```
#[verifier::external_body]
struct MySeq;

impl MySeq {
  spec fn empty() -> MySeq;
  spec fn len(self) -> nat;
  spec fn push(self, value: int) -> MySeq;
}

proof fn test(x: int, y: int) {
  let s = MySeq::empty();
  assert(s.push(x + y)
    == s.push(y + x));
}
```



$$x + y == y + x$$



$$\text{push}(s, x + y) == \text{push}(s, y + x)$$

Interpreted functions (nonrecursive)

Core Components (always enabled)

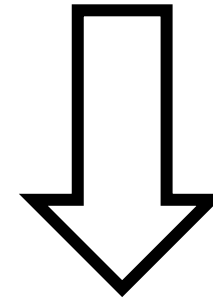
Uninterp funcs

```
spec fn f(x,...,x);  
f(e,...,e)
```

Quantifiers

```
forall|x: t| e  
exists|x: t| e
```

```
spec fn f(x: int, y: int) -> int {  
  x + 2 * y  
}
```



```
spec fn f(x: int, y: int) -> int;
```

axiom:

```
forall|x: int, y: int|  
  #[trigger] f(x, y) == x + 2 * y
```

Interpreted functions (recursive)

Core Components (always enabled)

Uninterp funcs

```
spec fn f(x,...,x);  
f(e,...,e)
```

Quantifiers

```
forall |x: t| e  
exists |x: t| e
```

```
spec fn f(x: int) -> bool  
  decreases x  
{  
  x <= 0 || !f(x - 1)  
}
```

Naive axiom generates matching loop:

```
f(x)  
== x <= 0 || !f(x - 1)  
== x <= 0 || !(x - 1 <= 0 || !f(x - 1 - 1))  
== x <= 0 || !(x - 1 <= 0 || !(x - 1 - 1 <= 0 || !f(...)))  
...
```

Verus axiom uses “fuel” to limit unrolling depth

```
proof fn test() {  
  reveal_with_fuel(f, 4);  
  assert(!f(3));  
}
```

Assert-by for bit vector, nonlinear arith

Core Components (always enabled)

SAT solver

true | false | !e
e && e | ...

Linear int arith

-2 | -1 | 0 | 1 | 2
| e + e | e - e
| e <= e | ...

Uninterp funcs

spec fn f(x,...,x);
f(e,...,e)

Datatypes

struct S { ... }
enum E { ... }
S(e,...,e) | e.f | ...

Quantifiers

forall | x: t | e
exists | x: t | e

Additional Components (via “assert by”)

Bit vector arith

e & e | e << e |
...

Nonlin int arith

e * e
e / e | e % e

```
proof fn test_by(x: u32)
  requires x <= 200,
{
  assert(x >> 8 == 0) by(bit_vector)
  requires(x <= 255);
}
```

Assert-by for bit vector, nonlinear arith

Core Components (always enabled)

SAT solver

true | false | !e
e && e | ...

Linear int arith

-2 | -1 | 0 | 1 | 2
| e + e | e - e
| e <= e | ...

Uninterp funcs

spec fn f(x,...,x);
f(e,...,e)

Datatypes

struct S { ... }
enum E { ... }
S(e,...,e) | e.f | ...

Quantifiers

forall | x: t | e
exists | x: t | e

Additional Components (via “assert by”)

Bit vector arith

e & e | e << e |
...

Nonlin int arith

e * e
e / e | e % e

```
proof fn test_by(x: u32, y: u32)
  requires x > 1 && y > 0
{
  assert(x * y > y) by(nonlinear_arith)
  requires(x > 1 && y > 0);
}
```

Quantifiers

Quantifiers

forall $|x: t| e$
exists $|x: t| e$

```
spec fn f(i: int) -> int;  
spec fn g(i: int) -> int;  
  
proof fn test()  
  requires  
    forall|x: int, y: int| f(x) > g(y)  
  ensures  
    f(g(7)) > g(3)  
{  
}
```

Quantifiers

Quantifiers

forall $|x: t| e$
exists $|x: t| e$

Example:

$\forall f, g.$

(forall $|x: \text{int}, y: \text{int}| f(x) > g(y)$)

$\implies f(g(7)) > g(3)$

Quantifiers

Quantifiers

forall |x: t| e
exists |x: t| e

Example (as satisfiability query):

$\exists f, g.$

(forall |x: int, y: int| f(x) > g(y))
&& f(g(7)) <= g(3)

Infinitely many possible instantiations of x, y

...

&& f(2) > g(7)

&& f(4) > g(6)

&& f(4) > g(7)

&& f(5) > g(10)

&& f(5) > g(-5)

...

Quantifiers

Quantifiers

forall |x: t| e
exists |x: t| e

Example:

$\exists f, g.$

pattern ("trigger") to match

(forall |x: int, y: int| #[trigger f(x), g(y)] f(x) > g(y))
&& f(g(7)) <= g(3)

Quantifiers

Quantifiers

forall |x: t| e
exists |x: t| e

Example: (alternate trigger notation for trigger $f(x)$, $g(y)$)

$\exists f, g.$

(forall |x: int, y: int| $\#[\text{trigger}] f(x) > \#[\text{trigger}] g(y)$)
&& $f(g(7)) \leq g(3)$

Quantifiers

Quantifiers

forall |x: t| e
exists |x: t| e

Example:

$\exists f, g.$

pattern ("trigger") to match

(forall |x: int, y: int| #![trigger f(x), g(y)] f(x) > g(y))

&& f(g(7)) <= g(3)

matches
f(x) with
x = g(7)

matches
g(y) with
y = 3

Quantifiers

Quantifiers

forall $|x: t| e$
exists $|x: t| e$

Example:

$\exists f, g.$

pattern ("trigger") to match

(forall $|x: \text{int}, y: \text{int}|$ $\text{\#!}[\text{trigger } f(x), g(y)] f(x) > g(y)$)
&& $f(g(7)) \leq g(3)$ && $f(g(7)) > g(3)$

matches
 $f(x)$ with
 $x = g(7)$

matches
 $g(y)$ with
 $y = 3$

contradiction
(no counterexample,
original formula is valid)

Quantifiers

Quantifiers

forall |x: t| e
exists |x: t| e

Example:

$\exists f, g.$

pattern ("trigger") to match

(forall |x: int, y: int| #[trigger f(x), g(y)] f(x) > g(y))
&& f(g(7)) <= g(3)

Triggers may be written by user,
or chosen automatically by Verus.

Quantifiers

Quantifiers

forall | x: t | e
exists | x: t | e

Example:

$\exists f, g.$

pattern ("trigger") to match

(forall | x: int, y: int | #[trigger f(x), g(y)] f(x) > g(y))

&& f(g(7)) <= g(3)

- The trigger must mention all the quantified variables
 - both x and y
- The trigger matches only if **all** of its patterns match
 - both f(x) and g(y) must match some expressions

Quantifiers

Quantifiers

forall $|x: t| e$
exists $|x: t| e$

Example: *badly behaved trigger*

$\exists f.$

(forall $|x: \text{int}|$ *#[trigger f(x)]* $f(f(x)) > f(x)$)

&& $f(3) \leq f(f(2))$

&& $f(f(3)) > f(3)$

&& $f(f(f(3))) > f(f(3))$

&& $f(f(f(f(3)))) > f(f(f(3)))$

...

Beware of infinite matching loops

Quantifiers

Quantifiers

forall $|x: t| e$
exists $|x: t| e$

Example:

nicely behaved trigger

$\exists f.$

(forall $|x: \text{int}|$ *#[trigger f(f(x))]* $f(f(x)) > f(x)$)

&& $f(3) \leq f(f(2))$

&& $f(f(2)) > f(2)$

Triggers are the programming language for quantifiers.
Choose triggers carefully!

Higher-order features,
vstd library

Higher-order features: spec_fn

```
proof fn test() {  
  let f: spec_fn(int, int) -> int = |x: int, y: int| x + x - y;  
  let g: spec_fn(int, int) -> int = |x: int, y: int| 2 * x - y;  
  assert(f(30, 20) == 40);  
  assert(forall |x: int, y: int| #[trigger] f(x, y) == g(x, y));  
}
```

SMT encoding (simplified):

```
(declare-sort Fun)  
(declare-const f Fun)  
(declare-const g Fun)  
(declare-fun apply (Fun Int Int) Int)  
(assert (forall ((x Int) (y Int)) (= (apply f x y) (Sub (Add x x) y))))  
(assert (forall ((x Int) (y Int)) (= (apply g x y) (Sub (Mul 2 x) y))))
```

Note: real SMT encoding supports various types/arities and supports direct equality of similar closures

Verus's vstd library: Seq, Set, Map, etc.

// finite sequence

```
struct Seq<A> { ... }
```

```
impl<A> Seq<A> {
```

```
    spec fn empty() -> Seq<A>;
```

```
    spec fn new(len: nat, f: spec_fn(int) -> A) -> Seq<A>;
```

```
    spec fn len(self) -> nat;
```

```
    spec fn index(self, i: int) -> A;
```

```
    spec fn push(self, a: A) -> Seq<A>;
```

```
    ...
```

```
}
```

```
proof fn test() {
```

```
    let s: Seq<int> = Seq::new(5, |i: int| 10 * i);
```

```
    assert(s.len() == 5);
```

```
    assert(s.index(3) == 30); // syntax sugar: s[3] == 30
```

```
    let s2 = s.push(50);
```

```
    assert(s2.len() == 6);
```

```
}
```

*Rust feature note:
"impl" and "self" are for
namespacing and "." syntax*

// finite or infinite set

```
struct Set<A> { set: spec_fn(A) -> bool }
```

```
impl<A> Set<A> {
```

```
    spec fn empty() -> Set<A> { Set { set: |a: A| false } }
```

```
    spec fn new(f: spec_fn(A) -> bool) -> Set<A> { Set { set: f } }
```

```
    spec fn contains(self, a: A) -> bool { (self.set)(a) }
```

```
    spec fn insert(self, a: A) -> Set<A> { ... }
```

```
    ...
```

```
}
```

// finite or infinite map

```
struct Map<K, V> { mapping: spec_fn(K) -> Option<V> }
```

```
impl<K, V> Map<K, V> {
```

```
    spec fn empty() -> Map<K, V> { Map { mapping: |k| None } }
```

```
    spec fn insert(self, key: K, value: V) -> Map<K, V> { ... }
```

```
    ...
```

```
}
```

Axioms and lemmas via function calls

```
proof fn axiom_seq_push_len<A>(s: Seq<A>, a: A)
  ensures
    s.push(a).len() == s.len() + 1
{ ... }
```

```
proof fn test() {
  let s: Seq<int> = Seq::new(5, |i: int| 10 * i);
  let s2 = s.push(50);
  axiom_seq_push_len(s, 50);
  assert(s2.len() == 6);
}
```

Axioms and lemmas via broadcast

```
broadcast proof fn axiom_seq_push_len<A>(s: Seq<A>, a: A)
  ensures
    #[trigger] s.push(a).len() == s.len() + 1
{ ... }
```

```
proof fn test() {
  broadcast use axiom_seq_push_len;
```

```
  let s: Seq<int> = Seq::new(5, |i: int| 10 * i);
  let s2 = s.push(50);
  assert(s2.len() == 6);
}
```



forall | s: Seq<A>, a: A | **#!**[trigger s.push(a).len()]
s.push(a).len() == s.len() + 1

Broadcast groups and defaults

```
struct Seq<A> { ... }  
impl<A> Seq<A> { ... }  
broadcast proof fn axiom_seq_push_len<A>(...) ...  
...
```

```
broadcast group group_seq_axioms {  
  axiom_seq_push_len,  
  axiom_seq_push_index_same,  
  axiom_seq_push_index_different,  
  ...  
}
```

```
#[verifier::broadcast_use_by_default_when_this_crate_is_imported]  
broadcast group group_vstd_default {  
  group_seq_axioms,  
  group_set_axioms,  
  group_map_axioms,  
  ...  
}
```

Verus vstd crate

import vstd crate

user crate

```
use vstd::prelude::*;  
  
proof fn test() {  
  let s: Seq<int> = Seq::new(5, |i: int| 10 * i);  
  let s2 = s.push(50);  
  assert(s2.len() == 6);  
}
```

Equality via extensionality

```
proof fn test() {  
  let s1 = seq![10] + seq![20, 30];  
  let s2 = seq![10, 20] + seq![30];  
  // Does SMT solver know s1 == s2?  
  
}
```

Seq::add([10], [20, 30]) ==
Seq::add([10, 20], [30])

Uninterp funcs

spec fn f(x,...,x);
f(e,...,e)

congruence
doesn't
help here!

[10] == [10, 20]
[20, 30] == [30]

Equality via extensionality

```
broadcast proof fn axiom_seq_ext_equal<A>(s1: Seq<A>, s2: Seq<A>)
```

```
  ensures
```

```
    #[trigger] (s1 ==~ s2) <==>
```

```
      s1.len() == s2.len() && (forall |i: int| 0 <= i < s1.len() ==> s1[i] == s2[i])
```

```
{ ... }
```

```
proof fn test() {
```

```
  let s1 = seq![10] + seq![20, 30];
```

```
  let s2 = seq![10, 20] + seq![30];
```

```
  // Does SMT solver know s1 == s2?
```

```
  // ==~ is semantically equivalent to ==, but will trigger additional lemmas
```

```
  assert(s1 ==~ s2); // triggers axiom_seq_ext_equal
```

```
  // Now SMT solver knows s1 == s2
```

```
}
```

```
// Note: Verus automatically replaces == with ==~ in assert, ensures, invariant
```

Decidable and undecidable theories

DECIDABLE THEORIES

boolean expressions

$e ::= \text{true} \mid \text{false} \mid !e \mid e \ \&\& \ e \mid e \implies e \mid \dots$

linear integer arithmetic

$e ::= \dots \mid -2 \mid -1 \mid 0 \mid 1 \mid 2 \mid e + e \mid e - e \mid e \leq e \mid \dots$

bit vector arithmetic

$e ::= \dots \mid e \ \& \ e \mid e \ll e \mid \dots$

uninterpreted functions

$e ::= \dots \mid f(e, \dots, e)$

datatypes

$e ::= \dots \mid S(e, \dots, e) \mid S.f \mid \text{match } e \{ \dots \}$

UNDECIDABLE

nonlinear integer arithmetic

$e ::= \dots \mid e * e \mid e / e \mid e \% e$

quantifiers

$e ::= \dots \mid (\text{forall } x: t \mid e) \mid (\text{exists } x: t \mid e)$

Verus uses quantifiers for many broadcast lemmas, including $\approx$,
so we can't expect completely automated proofs

Decidability, undecidability, automation

- For decidable theories, we can hope for complete SMT automation
 - ... except when the SMT decision procedures are too slow
 - `assert(...)` by `bit_vector`
- But most Verus code relies on features beyond the decidable fragment
 - Nonlinear arithmetic: `assert(...)` by `nonlinear_arith`
 - Verus's lemmas for Seq, Set, Map, etc.
 - Equality via extensionality (`=~=`)
 - User-supplied quantifiers
 - `forall |i: int, j: int| 0 <= i <= j < v.len() ==> v[i] <= v[j]`
 - Recursive definitions and proofs by induction

So we get some mixture of automation and explicit proofs

Generics and traits (briefly)

Rust feature: generics

In AIR, SMT: x: Poly y: Poly

```
spec fn f<A>(x: A, y: A) -> A { x }

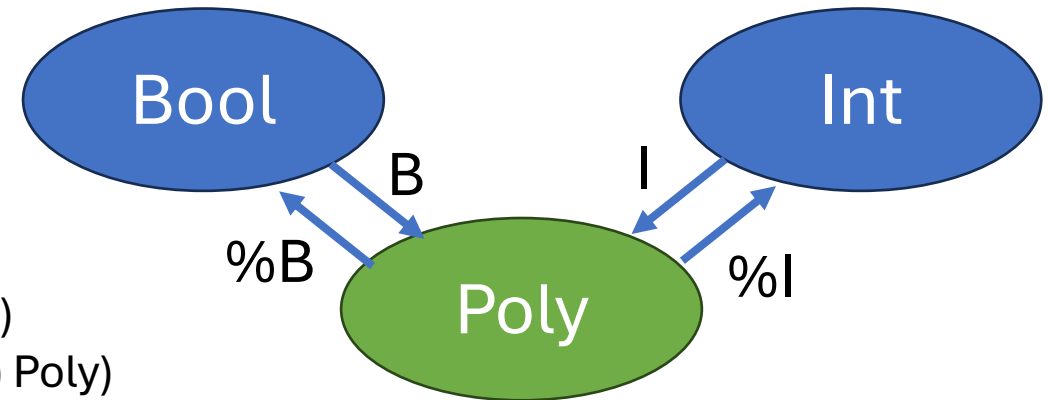
proof fn test1() {
  assert(f::<int>(10, 20) == 10);
  assert(f::<bool>(false, true) == false);
}

proof fn test2() {
  assert(f(10int, 20) == 10);
  assert(f(false, true) == false);
}
```

In AIR, SMT: (B false) (B true)

In AIR, SMT: (I 10) (I 20)

```
(declare-sort Poly)
(declare-fun I (Int) Poly)
(declare-fun B (Bool) Poly)
(declare-fun %I (Poly) Int)
(declare-fun %B (Poly) Bool)
(axiom (forall ((x Bool)) (= x (%B (B x)))))
(axiom (forall ((x Int)) (= x (%I (I x)))))
... more axioms ...
```



Rust feature: traits (aka Haskell type classes)

```
trait Clone {  
    fn clone(&self) -> Self;  
}  
  
trait Copy: Clone { }  
  
trait Default {  
    fn default() -> Self;  
}  
  
trait Add<Rhs> {  
    type Output;  
    fn add(self, rhs: Rhs) -> Self::Output;  
}
```

```
struct Complex { real: f64, imag: f64 }  
impl Clone for Complex {  
    fn clone(&self) -> Complex {  
        Complex { real: self.real, imag: self.imag }  
    }  
}  
impl Default for Complex {  
    fn default() -> Complex {  
        Complex { real: 0.0, imag: 0.0 }  
    }  
}  
impl Add<Complex> for Complex {  
    type Output = Complex;  
    fn add(self, rhs: Complex) -> Complex {  
        Complex {  
            real: self.real + rhs.real,  
            imag: self.imag + rhs.imag,  
        }  
    }  
}
```

```
fn f<T>(x: T) -> T::Output  
where  
    T: Default,  
    T: Add<T>,  
{  
    x + T::default()  
}  
fn test() {  
    let c1 = Complex {  
        real: 3.14,  
        imag: 2.72,  
    };  
    let c2 = Complex::default();  
    let c3 = c1.clone();  
    let c4 = c1 + c2 + c3;  
    let c5 = f(c4);  
}
```

Notes:

- the "self" parameter uses "." syntax
- "impl S for Trait { ... }" is a trait impl
- "impl S { ... }" is for namespacing and "." syntax, unrelated to traits

Verus View trait: exec value -> spec value

```
trait View {  
  type V;  
  spec fn view(&self) -> Self::V;  
}  
  
impl View for String {  
  type V = Seq<char>;  
  spec fn view(&self) -> Seq<char> { ... }  
}  
  
impl<T> View for Vec<T> {  
  type V = Seq<T>;  
  spec fn view(&self) -> Seq<T> { ... }  
}  
  
impl<T> View for HashSet<T> {  
  type V = Set<T>;  
  spec fn view(&self) -> Set<T> { ... }  
}  
  
impl<Key, Value> View for HashMap<Key, Value> {  
  type V = Map<Key, Value>;  
  spec fn view(&self) -> Map<Key, Value> { ... }  
}
```

```
fn test() {  
  let mut v: Vec<u8> = Vec::new();  
  v.push(10);  
  v.push(20);  
  v.push(30);  
  proof {  
    let s: Seq<u8> = v@;  
    assert(s == seq![10, 20, 30]);  
  }  
}
```

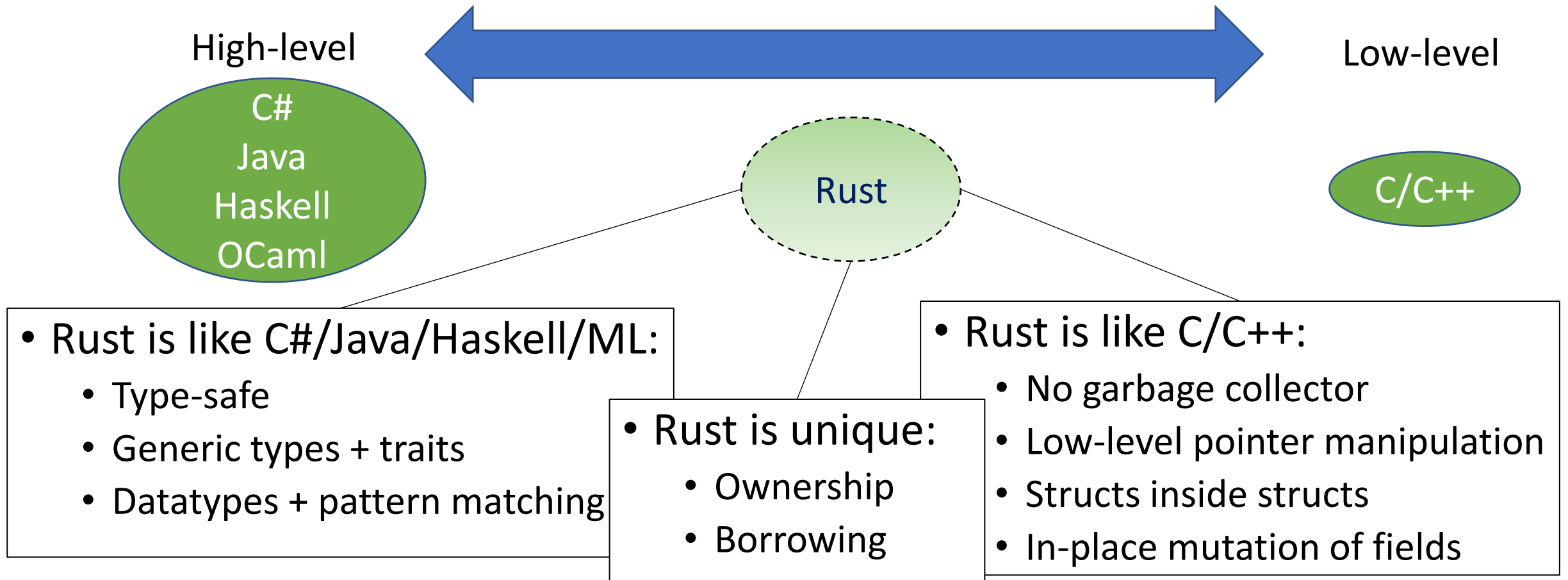
“v@” is syntax sugar for “v.view()”

```
fn test() {  
  let mut v: Vec<u8> = Vec::new();  
  v.push(10);  
  v.push(20);  
  v.push(30);  
  let ghost s: Seq<u8> = v@;  
  assert(s == seq![10, 20, 30]);  
}
```

“ghost”: proof-only variable

Ownership, &, &mut

Why write systems code in Rust?



Ownership in Rust

```
fn consume(v: Vec<u8>) {  
}  
  
fn test1() {  
    let mut v = Vec::new();  
    v.push(10);  
    consume(v);  
}  
  
fn test2() {  
    let mut v = Vec::new();  
    consume(v);  
    v.push(10); // ERROR: v no longer available  
}
```

Note: types implementing Copy (bool, u8, ...) are copied, not moved
Vec does *not* implement Copy, so it is moved, not copied

Borrowing in Rust

```
fn borrow(v: &Vec<u8>) {  
}  
  
fn test1() {  
    let mut v = Vec::new();  
    v.push(10);  
    borrow(&v);  
}  
  
fn test2() {  
    let mut v = Vec::new();  
    borrow(&v);  
    v.push(10); // ok  
}
```

```
fn borrow(v: &mut Vec<u8>) {  
}  
  
fn test1() {  
    let mut v = Vec::new();  
    v.push(10);  
    borrow(&mut v);  
}  
  
fn test2() {  
    let mut v = Vec::new();  
    borrow(&mut v);  
    v.push(10); // ok  
}
```

Borrowing in Rust

```
fn borrow(  
    v1: &Vec<u8>,  
    v2: &Vec<u8>,  
) {  
}  
  
fn test() {  
    let mut v = Vec::new();  
    borrow(&v, &v); // ok  
}
```

```
fn borrow(  
    v1: &Vec<u8>,  
    v2: &mut Vec<u8>,  
) {  
}  
  
fn test() {  
    let mut v = Vec::new();  
    borrow(&v, &mut v); // error  
}
```

```
fn borrow(  
    v1: &mut Vec<u8>,  
    v2: &mut Vec<u8>,  
) {  
}  
  
fn test() {  
    let mut v = Vec::new();  
    borrow(&mut v, &mut v); // error  
}
```

multiple shared (immutable) references to v simultaneously (aliasing ok),
or
one mutable reference allowed to v (no aliasing)

Note: &t implements Copy, while &mut t does *not* implement Copy

Mutation in Rust

```
fn f(v: Vec<u8>) {  
    v.push(10); // ok  
}
```

```
fn f(v: &Vec<u8>) {  
    v.push(10); // error  
}
```

```
fn f(v: &mut Vec<u8>) {  
    v.push(10); // ok  
}
```

- mutation allowed on owned value and mutable reference
- shared references does not allow mutation

So Rust allows ***either mutation or aliasing*** of a value,
but not both mutation and aliasing together.

This is an unusual discipline.

Most languages (C, C++, Java, C#, ML, JavaScript, Python, Dafny, ...) allow mutation and aliasing together.

Mutation + aliasing is annoying for verification

// Dafny code

method M(x: array<int>, y: array<int>)

requires x.Length > 0

requires y.Length > 0

requires y[0] == 3

modifies x

{

assert y[0] == 3;

x[0] := 5;

assert y[0] == 3; // error

}

// Dafny code

method M(x: array<int>, y: array<int>)

requires x != y

requires x.Length > 0

requires y.Length > 0

requires y[0] == 3

modifies x

{

assert y[0] == 3;

x[0] := 5;

assert y[0] == 3; // ok

}

// Rust/Verus

fn f(x: &mut Vec<u8>, y: &Vec<u8>)

requires

x.len() > 0,

y.len() > 0,

y[0] == 3,

{

assert(y[0] == 3);

x[0] = 5;

assert(y[0] == 3); // ok

}

Classical verification (e.g. Dafny): one big heap

```
// Dafny code
```

```
method M(x: array<int>, y: array<int>)  
  requires x != y
```

Classical approach:

- x and y are addresses
- objects (e.g. arrays) live in heap
- need to worry about aliasing

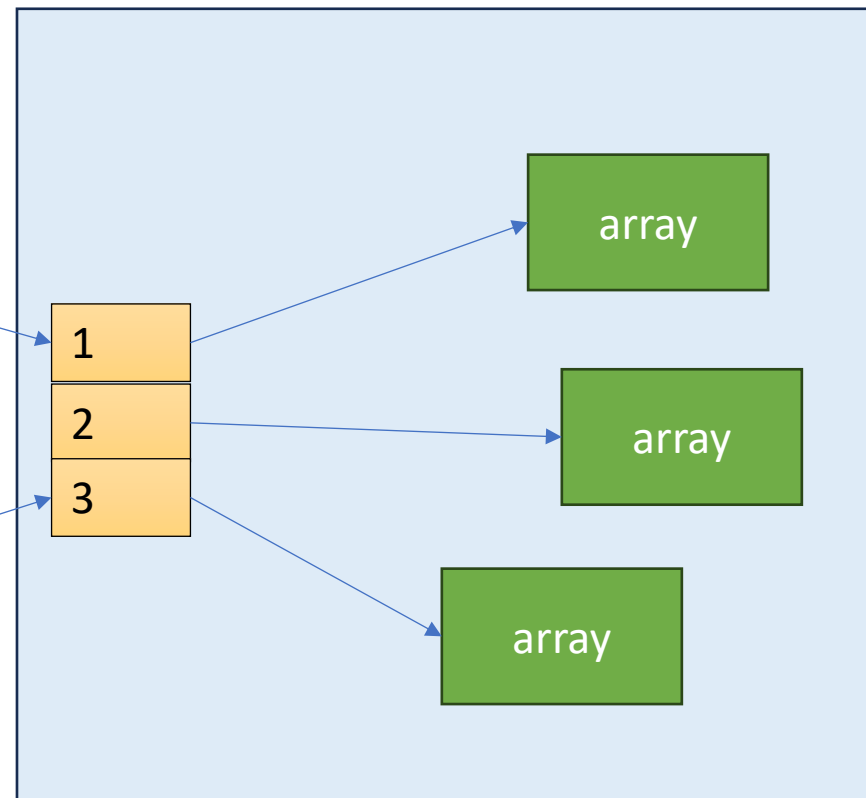
(declare-const y! Int)

y = 1

(declare-const x! Int)

x = 3

(declare-var heap@ MapIntToObject)



Rust verification: no heap needed

In AIR/SMT encoding, values are stored in the references, not in a separate heap:

- SMT representation of `&t` is the same as SMT representation of `t`
- SMT representation of `&mut t` is a pair of `t` and extra information

Let $[[t]]$ be the SMT representation of Rust type `t`:

$[[\text{bool}]] = \text{Bool}$

$[[\text{int}]] = \text{Int}$

$[[\&t]] = [[t]]$

$[[\&\text{mut } t]] = ([[t]], \dots \text{extra-stuff} \dots)$

$[[\text{Complex}]] = \text{Complex}$

...

```
struct Complex { real: f64, imag: f64 }
```

```
fn f(x: Complex, y: &Complex, z: &mut Complex) { ... }
```

(declare-const x! Complex)

(declare-const y! Complex)

(declare-const z! Poly)

Rust &mut is powerful and complex

```
struct S { a: u32, b: u32, c: u32 }
```

```
fn f(uref: &mut u32) {  
    *uref += 1;  
}  
  
fn g() {  
    let mut s = S { a: 10, b: 20, c: 30 };  
    f(&mut s.c);  
    assert!(s == S { a: 10, b: 20, c: 31 });  
}
```

```
fn f(sref: &mut S) -> &mut u32 {  
    &mut sref.c  
}  
  
fn g() {  
    let mut s = S { a: 10, b: 20, c: 30 };  
    let p = f(&mut s);  
    *p = *p + 1;  
    assert!(s == S { a: 10, b: 20, c: 31 });  
}
```

```
fn f(flag: bool, sref: &mut S) -> &mut u32 {  
    if flag { &mut sref.b } else { &mut sref.c }  
}  
  
fn g() {  
    let mut s = S { a: 10, b: 20, c: 30 };  
    let p = f(true, &mut s);  
    *p = *p + 1;  
    assert!(s == S { a: 10, b: 21, c: 30 });  
}
```

[[&mut t]] = ([[t]], ...extra-stuff...)

The “extra-stuff” says how to put the new [[t]] into the place the old [[t]] was borrowed from

SMT encoding of &mut

$[[\&\text{mut } t]] = ([[t]], \dots \text{extra-stuff} \dots)$

The “extra-stuff” says how to put the new $[[t]]$ into the place the old $[[t]]$ was borrowed from

```
fn f(uref: &mut u32) {  
    *uref += 1;  
}  
  
fn g() {  
    let mut s = S { a: 10, b: 20, c: 30 };  
    f(&mut s.c);  
    assert!(s == S { a: 10, b: 20, c: 31 });  
}
```

Lots of research in last 10 years, various approaches

- “extra-stuff” could be address (but then we’d need a heap, which we don’t want)
- “extra-stuff” could be a function generating new S from new $[[t]]$
- “extra-stuff” could be a function saying what’s true about new S , given new $[[t]]$

Verus (following RustHorn, Creusot) uses “prophecies”:

- “RustHorn: CHC-based Verification for Rust Programs”
(Matsushita, Tsukada, and Kobayashi, ESOP 2020)
“extra-stuff” is a “prophecy” of type $[[t]]$ naming the final value of $[[t]]$

$[[\&\text{mut } t]] = (\quad [[t]], \quad [[t]] \quad)$
(current value, prophesized final value)

SMT encoding of &mut

$[[\&\text{mut } t]] = (\quad [[t]], \quad [[t]] \quad)$
(current value, prophesized final value)

```
fn f(uref: &mut u32) {  
    *uref += 1;  
}  
  
fn g() {  
    let mut s = S { a: 10, b: 20, c: 30 };  
    f(&mut s.c);  
    assert!(s == S { a: 10, b: 20, c: 31 });  
}
```

$s = S \{ a: 10, b: 20, c: 30 \}$

(declare-const X Int)

**Let X be the name of the prophesized final value of &mut s.c
Immediately replace old value (30) with X
 $s = S \{ a: 10, b: 20, c: X \}$**

$uref = (30, X)$

(assign s (... (c X)...))

$uref = (31, X)$

**At the end of f, the &mut expires (it is owned by f, and &mut is not Copy)
When the &mut expires, we know its final value: $X = 31$
 $uref = (31, 31)$**

**Knowing $X = 31$ allows g to conclude:
 $s = S \{ a: 10, b: 20, c: 31 \}$**

(assume (= (...prophecy...) (...current...)))

SMT encoding of &mut

$[[\&\text{mut } t]] = (\quad [[t]], \quad [[t]] \quad)$
(current value, prophesized final value)

```
fn f(uref: &mut u32)
  requires
    *uref < u32::MAX,
  ensures
    *final(uref) == *old(uref) + 1,
  {
    *uref += 1;
  }

fn g() {
  let mut s = S { a: 10, b: 20, c: 30 };
  f(&mut s.c);
  assert!(s == S { a: 10, b: 20, c: 31 });
}
```

$s = S \{ a: 10, b: 20, c: 30 \}$

Let X be the name of the prophesized final value of $\&\text{mut } s.c$
Immediately replace old value (30) with X
 $s = S \{ a: 10, b: 20, c: X \}$

$\text{uref} = (\text{30}, X)$

$\text{uref} = (31, X)$

At the end of f , the $\&\text{mut}$ expires (it is owned by f , and $\&\text{mut}$ is not Copy)
When the $\&\text{mut}$ expires, we know its final value: $X = 31$
 $\text{uref} = (31, 31)$

Knowing $X = 31$ allows g to conclude:
 $s = S \{ a: 10, b: 20, c: 31 \}$

SMT encoding of &mut

$[[\&\text{mut } t]] = (\quad [[t]], \quad [[t]] \quad)$
(current value, prophesized)

```
fn f(sref: &mut S) -> (r: &mut u32) {  
  &mut sref.c  
}
```

```
fn g() {  
  let mut s = S { a: 10, b: 20, c: 30 };  
  let p = f(&mut s);  
  *p = *p + 1;  
  assert!(s == S { a: 10, b: 20, c: 31 });  
}
```

$s = S \{ a: 10, b: 20, c: 30 \}$

Let X be the name of the prophesized final value of &mut s
Immediately replace old value (S { a: 10, b: 20, c: 30}) with X
 $s = X$

$sref = (S \{ a: 10, b: 20, c: 30 \}, X)$

Let Y be the name of the prophesized final value of &mut sref.c
 $sref = (S \{ a: 10, b: 20, c: Y \}, X)$
 $r = (30, Y)$
At the end of f, sref expires with $X = S \{ a: 10, b: 20, c: Y \}$

$p = (30, Y)$

$p = (31, Y)$

p expires with $Y = 31$

Knowing $X = S \{ a: 10, b: 20, c: Y \}$ and $Y = 31$
allows g to conclude: $s = S \{ a: 10, b: 20, c: 31 \}$

SMT encoding of &mut

$[[\&\text{mut } t]] = (\quad [[t]], \quad [[t]] \quad)$
(current value, prophesized)

```
fn f(sref: &mut S) -> (r: &mut u32)
  ensures
    *r == old(sref).c,
    final(sref).a == old(sref).a,
    final(sref).b == old(sref).b,
    final(sref).c == *final(r),
  {
    &mut sref.c
  }

fn g() {
  let mut s = S { a: 10, b: 20, c: 30 };
  let p = f(&mut s);
  *p = *p + 1;
  assert!(s == S { a: 10, b: 20, c: 31 });
}
```

$s = S \{ a: 10, b: 20, c: 30 \}$

Let X be the name of the prophesized final value of &mut s
Immediately replace old value ($S \{ a: 10, b: 20, c: 30 \}$) with X
 $s = X$

$sref = (S \{ a: 10, b: 20, c: 30 \}, X)$

Let Y be the name of the prophesized final value of &mut sref.c
 $sref = (S \{ a: 10, b: 20, c: Y \}, X)$
 $r = (30, Y)$

At the end of f, sref expires with $X = S \{ a: 10, b: 20, c: Y \}$

$p = (30, Y)$

$p = (31, Y)$

p expires with $Y = 31$

Knowing $X = S \{ a: 10, b: 20, c: Y \}$ and $Y = 31$
allows g to conclude: $s = S \{ a: 10, b: 20, c: 31 \}$

SMT encoding of &mut

$[[\&\text{mut } t]] = (\quad [[t]], \quad [[t]] \quad)$
(current value, prophesized final value)

```
fn f(flag: bool, sref: &mut S) -> (r: &mut u32)
  ensures
    *r == if flag { old(sref).b } else { old(sref).c },
    final(sref).a == old(sref).a,
    final(sref).b == if flag { *final(r) } else { old(sref).b },
    final(sref).c == if !flag { *final(r) } else { old(sref).c },
  {
    if flag { &mut sref.b } else { &mut sref.c }
  }

fn g() {
  let mut s = S { a: 10, b: 20, c: 30 };
  let p = f(true, &mut s);
  *p = *p + 1;
  assert(s == S { a: 10, b: 21, c: 30 });
}
```

Prophecy and soundness

Naive rules for prophecy can lead to unsoundness.

Example:

Let X be the name of a prophesized variable:

(declare-const X Int)

Resolve X to the expression (Add X 1)

(assume (= X (Add X 1)))

contradiction!

To prevent this, Verus limits use of prophetic variables to a few places like requires, ensures, assert, and specially marked prophetic spec fns

Verifying Unsafe Rust code

Separation logic / Rust ownership

Verus uses:

- Rust ownership to express ***separation*** (non-aliasing) of data
- SMT solvers (and higher-order extensions) to express ***logic***

“Separation logic” combines these and is the basis for much recent research in verification, particularly:

- unsafe code
- concurrent code (e.g. Iris)

Unfortunately, SMT solvers don’t support separation logic well (at least compared to classical logic). However, we can import separation logic ideas into Verus using Rust ownership and SMT-based logic.

Brief introduction to separation logic

Traditional classical logic applied to heaps $h = \{ \text{addr}_1 \rightarrow \text{data}_1, \dots, \text{addr}_n \rightarrow \text{data}_n \}$

Syntax: $\phi ::= \text{true} \mid \text{false} \mid \text{addr} \mapsto \text{value} \mid \phi \wedge \phi$

$h \models \text{true}$

$\{ \dots, \text{addr} \rightarrow \text{value}, \dots \} \models \text{addr} \mapsto \text{value}$

$(h \models \phi_1 \wedge \phi_2)$ iff $(h \models \phi_1 \text{ and } h \models \phi_2)$

```
fn f(x: addr, y: addr)
  requires x != y /\ x ↦ 100 /\ y ↦ 200
  { ... }
```

Separation logic applied to heaps $h = \{ \text{addr}_1 \rightarrow \text{data}_1, \dots, \text{addr}_n \rightarrow \text{data}_n \}$

Syntax: $\phi ::= \text{true} \mid \text{false} \mid \text{empty} \mid \text{addr} \mapsto \text{value} \mid \phi \wedge \phi \mid \phi * \phi$

$h \models \text{true}$

$\{ \} \models \text{empty}$

$\{ \text{addr} \rightarrow \text{value} \} \models \text{addr} \mapsto \text{value}$

$(h \models \phi_1 \wedge \phi_2)$ iff $(h \models \phi_1 \text{ and } h \models \phi_2)$

$(h \models \phi_1 * \phi_2)$ iff exists disjoint h_1, h_2 . $(h = h_1 \cup h_2 \text{ and } h_1 \models \phi_1 \text{ and } h_2 \models \phi_2)$

```
fn f(x: addr, y: addr)
  requires x ↦ 100 * y ↦ 200,
  { ... }
```

“*” tells us $x \neq y$

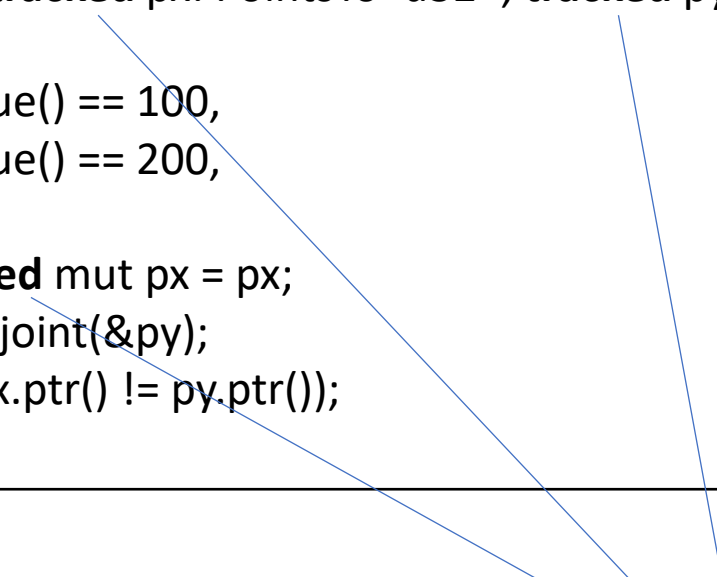
Separation Logic $\mapsto$ and $*$ is like “tracked PointsTo” and `,` in Verus

Separation logic pseudo-code

```
proof fn f(x: addr, y: addr)
  requires x  $\mapsto$  100 * y  $\mapsto$  200
{
  assert(x != y);
}
```

Actual Verus code for Rust unsafe raw pointers (using `vstd::raw_ptr::*`)

```
proof fn f(tracked px: PointsTo<u32>, tracked py: PointsTo<u32>)
  requires
    px.value() == 100,
    px.value() == 200,
{
  let tracked mut px = px;
  px.is_disjoint(&py);
  assert(px.ptr() != py.ptr());
}
```



tracked variables in proof code are:

- checked for ownership (and thus are “separated” like “ $*$ ”)
- unforgeable (can’t just make up an arbitrary value of any type `t`)

Separation Logic $\mapsto$ and $*$ is like “tracked PointsTo” and , in Verus

Separation logic pseudo-code

```
fn f(x: addr, y: addr)
  requires x  $\mapsto$  100 * y  $\mapsto$  200
{
  *x = *x + 1;
  assert(x  $\mapsto$  101);
}
```

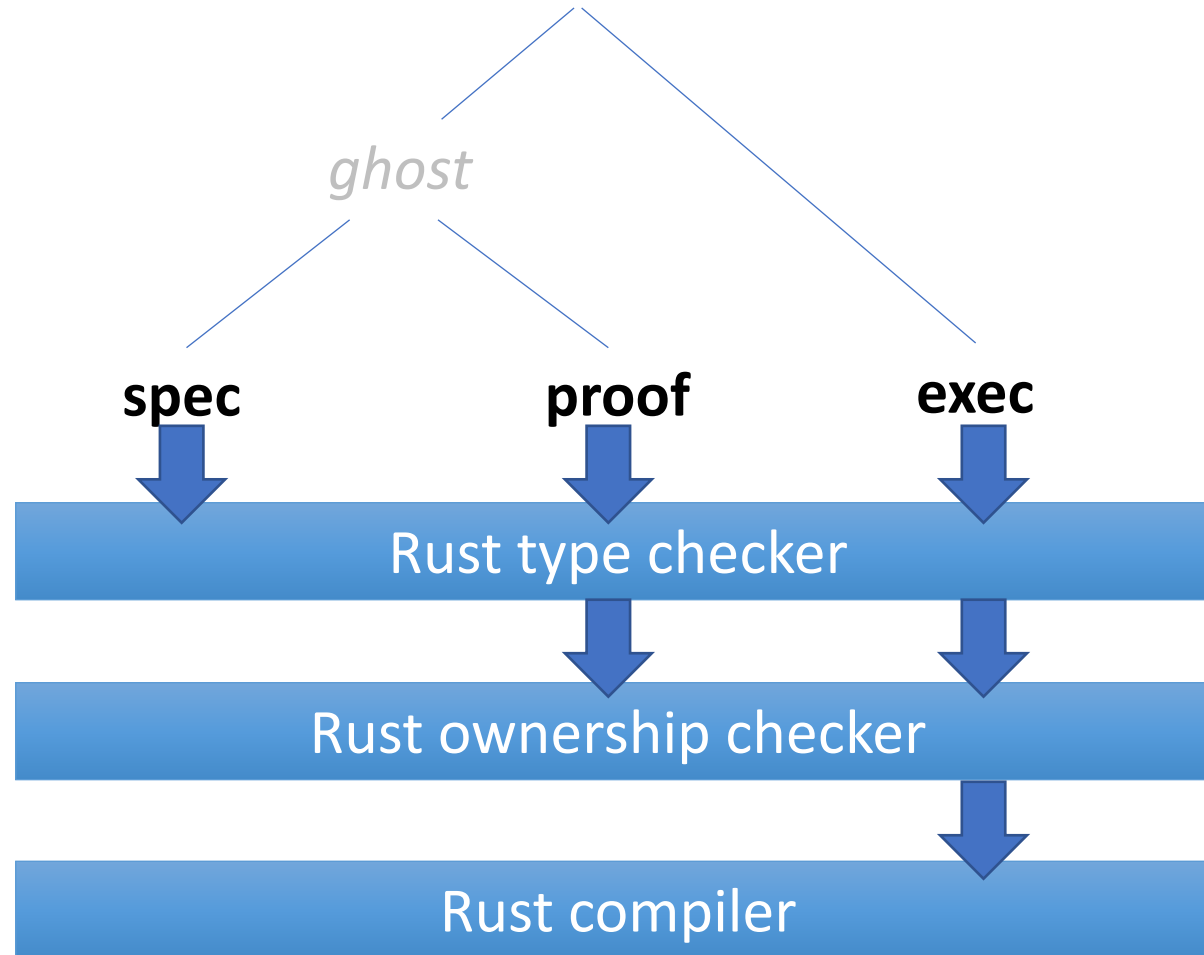
Rust raw pointer type “*mut t”
(like “&mut t”, but without
the safeguards)

Actual Verus code for Rust unsafe raw pointers (using vstd::raw_ptr::*)

```
fn f(x: *mut u32, Tracked(px): Tracked<PointsTo<u32>>,
    y: *mut u32, Tracked(py): Tracked<PointsTo<u32>>)
  requires
    px.value() == 100, px.ptr() == x, px.is_init(),
    py.value() == 200, py.ptr() == y, py.is_init(),
{
  let tracked mut px = px;
  let n = ptr_mut_read(x, Tracked(&mut px));
  ptr_mut_write(x, Tracked(&mut px), n + 1);
  assert(px.value() == 101);
}
```

tracked parameters and arguments in exec code must be wrapped in **Tracked(...)**
(ugly, but it’s a way to cram proofs into Rust’s existing type system)
Tracked is a “zero-size” type and is thus erased during compilation

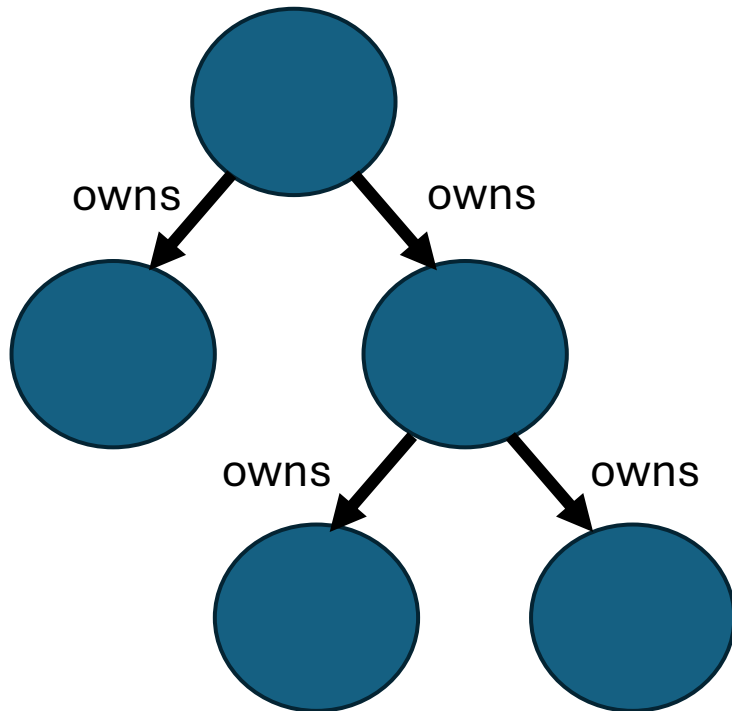
Ownership checking in proof and exec code



Verus can verify safety of “unsafe” Rust code

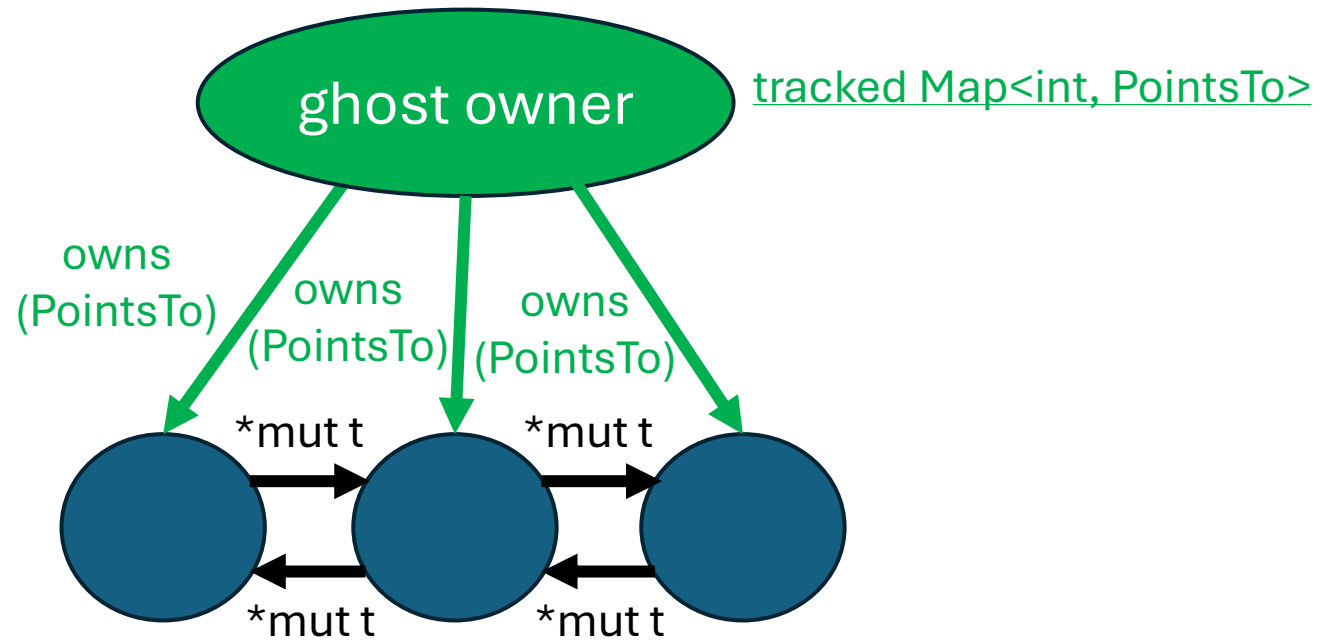
Trees:

Parent node "owns" child nodes;
Rust can check type safety



Doubly linked lists:

next and prev don't own each other,
so ordinary Rust can't check type safety.



Fractional permissions in separation logic

Split mutable points-to into multiple “fractional” immutable points-to

$$x \mapsto v \iff (x \mapsto^{0.5} v)^* (x \mapsto^{0.5} v) \iff (x \mapsto^{0.5} v)^* ((x \mapsto^{0.1} v)^* (x \mapsto^{0.4} v))$$

Each fractional permission has a real-number fraction between 0 and 1
Together, the fractions add up to 1.

Fractional permissions: Verus example

```
tracked struct FracGhost<T> { ... }
impl<T> FracGhost<T> {
  spec fn id(self) -> Loc;
  spec fn view(self) -> T;
  spec fn frac(self) -> real;

  proof fn new(v: T) -> (tracked r: FracGhost<T>)
    ensures r.frac() == 1.0real, r@ == v;

  proof fn split(tracked &mut self) -> (tracked r: Self)
    ensures
      final(self).frac() == old(self).frac() / 2.0real,
      r.frac() == old(self).frac() / 2.0real,
      ...;

  proof fn combine(tracked &mut self, tracked x: Self)
    requires old(self).id() == x.id(),
    ensures ..., final(self).frac() == old(self).frac() + x.frac(),
    ...
}
```

```
proof fn test() {
  let tracked mut x = FracGhost::new(100int);

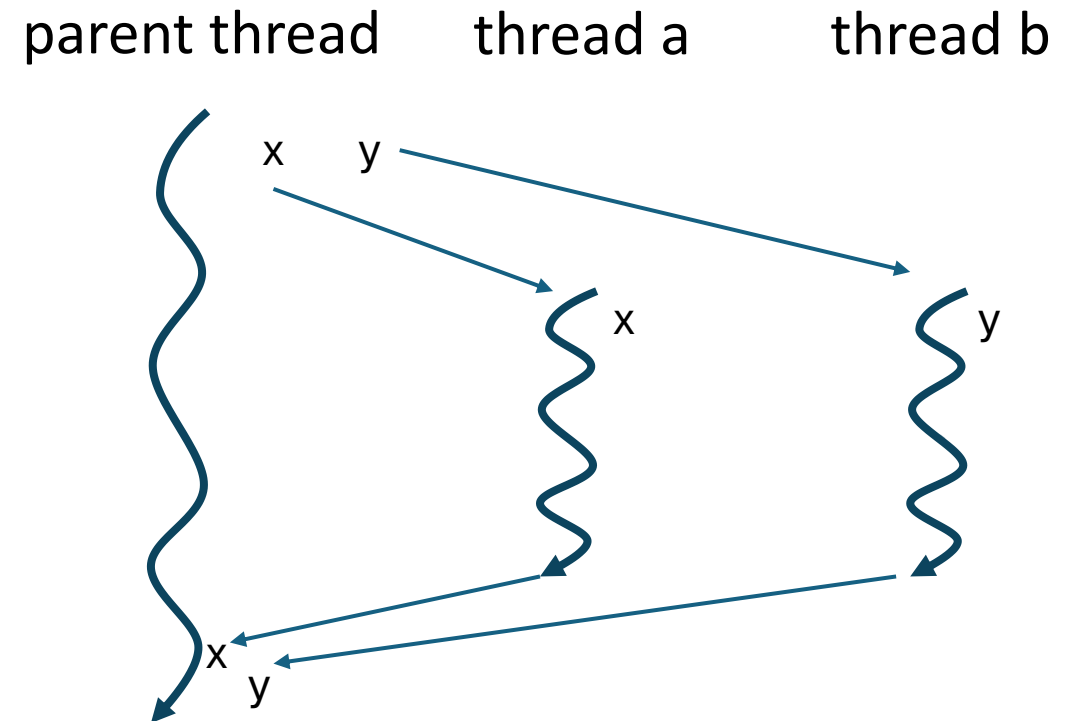
  let tracked y = x.split();
  assert(x.frac() == 0.5real);
  assert(y.frac() == 0.5real);
  assert(x@ == 100);
  assert(y@ == 100);

  x.combine(y);
  assert(x.frac() == 1.0real);
}
```

Concurrency

Concurrency via ownership/separation

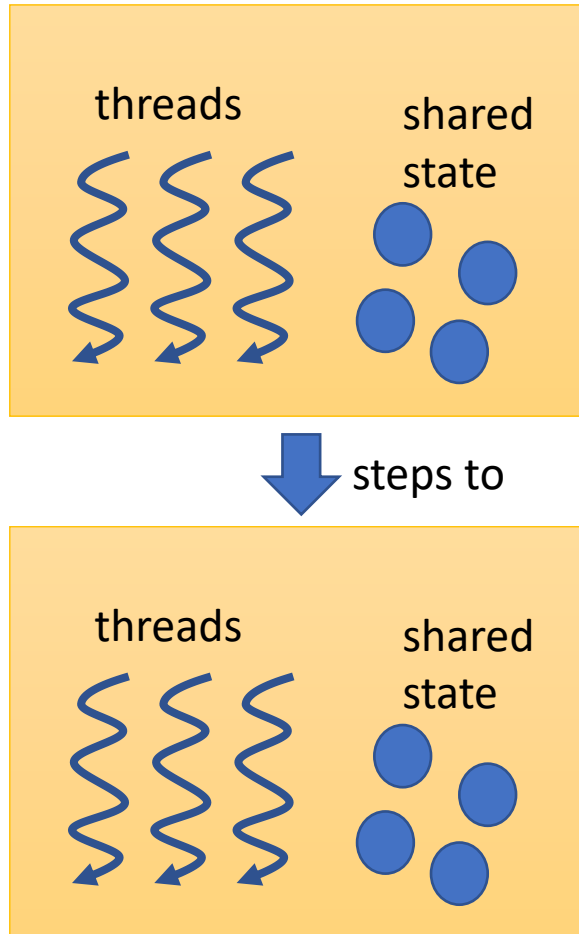
```
fn f(x: Vec<u8>, y: Vec<u8>) {  
    let thread_a = std::thread::spawn(|| {  
        let mut x = x;  
        x.push(10);  
        x  
    });  
  
    let thread_b = std::thread::spawn(|| {  
        let mut y = y;  
        y.push(20);  
        // let lx = x.len(); // error: x was moved into thread a  
        y  
    });  
  
    // let lx = x.len(); // error: x was moved into thread a  
    if let (Ok(x), Ok(y)) = (thread_a.join(), thread_b.join()) {  
        let lx = x.len();  
        let ly = y.len();  
    }  
}
```



*Concurrency is relatively easy when state is separate
But what if state is shared between threads?*

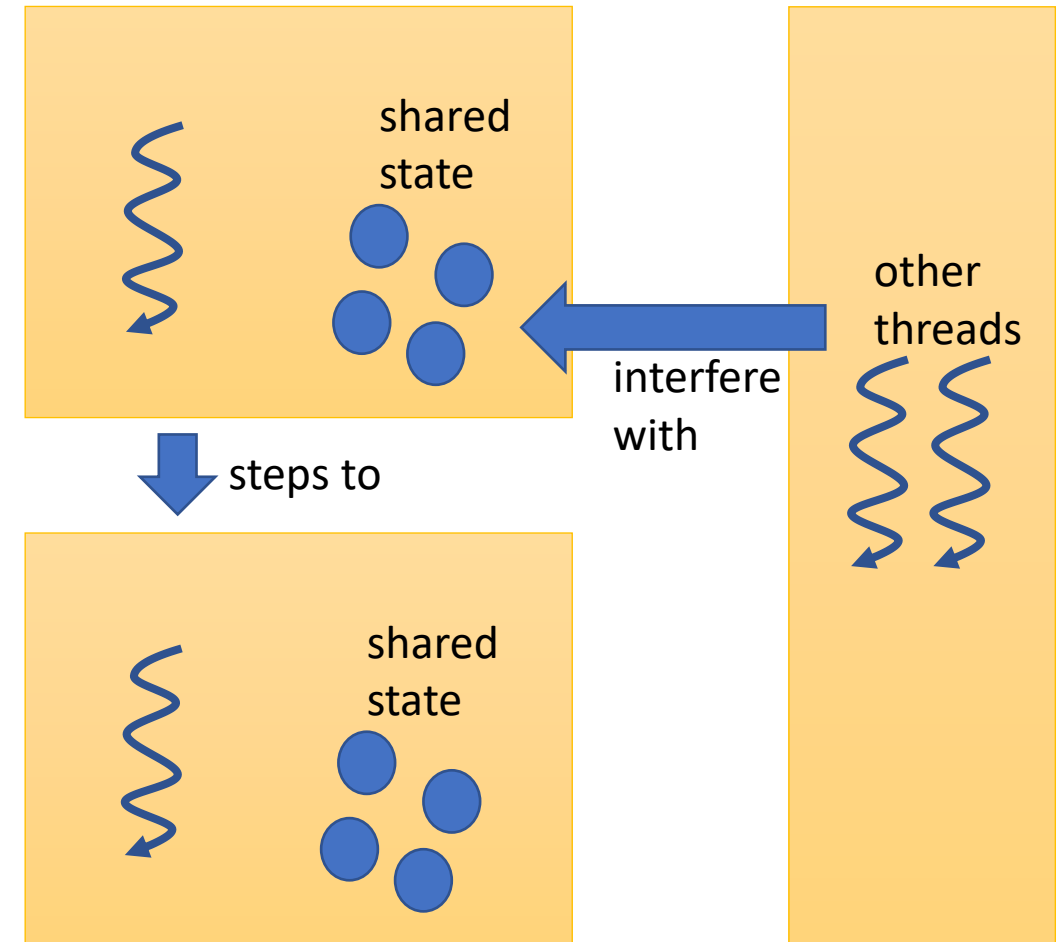
Concurrency with shared state

Perspective #1: whole system



Unwieldy for verifying large systems, though

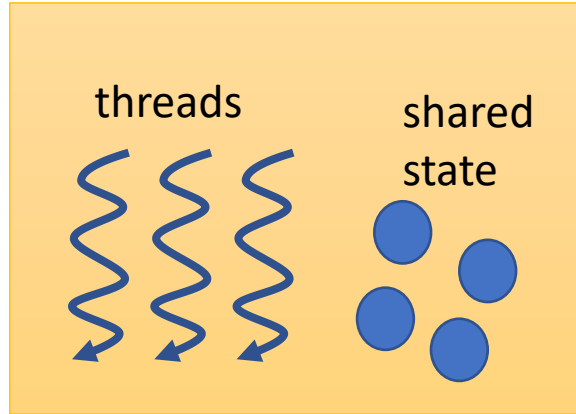
Perspective #2: one thread's view



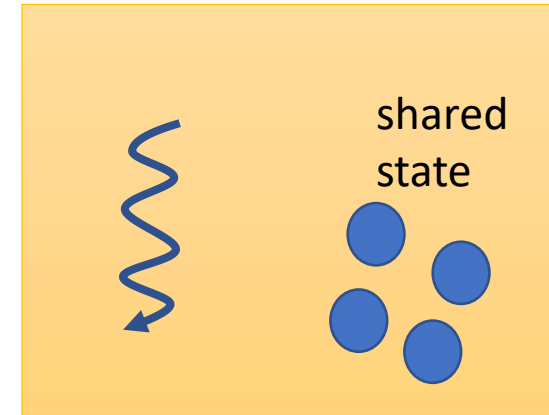
Example: Rely-Guarantee

Concurrency: global protocol, local actions

Perspective #1: whole system



Perspective #2: one thread's view



Describe an abstract protocol
as a whole system.

*Use invariants and/or state machines
to describe global protocol*

Describe each thread's implementation
of the protocol locally.

*Use ghost permissions, ownership
to perform local protocol steps*

Example: concurrent count-to-2

initial state: counter == 0

thread a



counter.fetch_add(1)

thread b



counter.fetch_add(1)

final state: counter == 2

Concurrent count-to-2, unverified Rust code

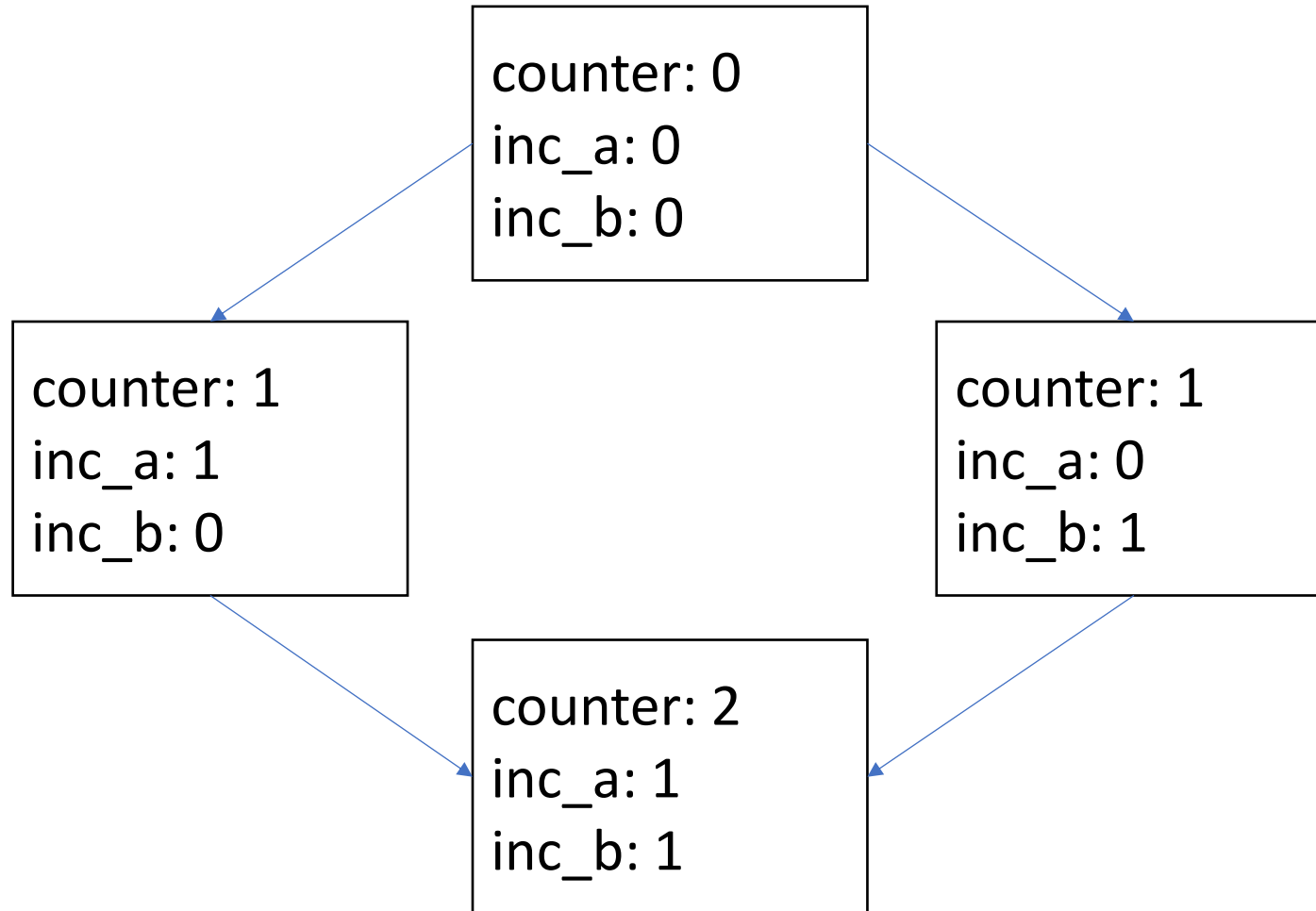
```
fn unverified_test() {  
    let ptr = AtomicU32::new(0);  
    let ptr = Arc::new(ptr);  
    let ptr_a = ptr.clone();  
    let ptr_b = ptr.clone();  
  
    let thread_a = std::thread::spawn(|| {  
        let ptr = ptr_a;  
        ptr.fetch_add(1, Ordering::SeqCst);  
    });  
  
    let thread_b = std::thread::spawn(|| {  
        let ptr = ptr_b;  
        ptr.fetch_add(1, Ordering::SeqCst);  
    });  
  
    if let (Ok(()), Ok(())) = (thread_a.join(), thread_b.join()) {  
        let n = ptr.load(Ordering::SeqCst);  
        assert!(n == 2);  
    }  
}
```

State machine for count-to-2

State type

```
counter: int  
inc_a: int  
inc_b: int
```

invariant:
 $\text{counter} = \text{inc_a} + \text{inc_b}$



State machine for count-to-2

State type

```
counter: int  
inc_a: int  
inc_b: int
```

Globally, we have a protocol (a state machine) with an invariant

Locally:

- thread_a is in charge of inc_a
- thread_b is in charge of inc_b

invariant:

$\text{counter} = \text{inc_a} + \text{inc_b}$

Verus supports two different approaches to express global/local cooperation:

- Individual separation-logic-style permissions (a la Iris)
- A dedicated macro-based state machine language

State machine for count-to-2

State type

```
counter: int  
inc_a: int  
inc_b: int
```

Globally, we have a protocol (a state machine) with an invariant
Locally:

- thread_a is in charge of inc_a
- thread_b is in charge of inc_b

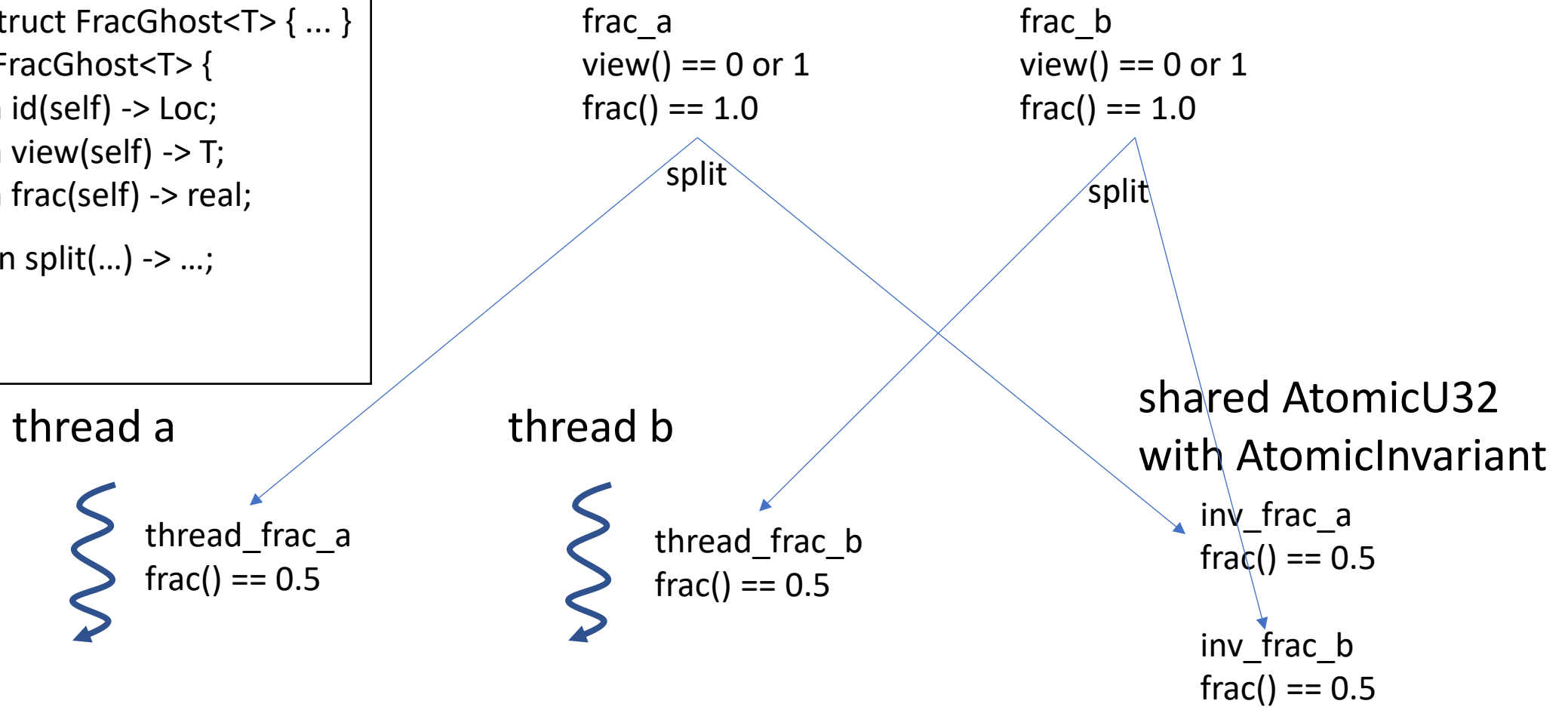
invariant:
 $\text{counter} = \text{inc_a} + \text{inc_b}$

Verus supports two different approaches to express global/local cooperation:

- **Individual separation-logic-style permissions (a la Iris)**
- A dedicated macro-based state machine language

Individual separation-logic-style permissions

```
tracked struct FracGhost<T> { ... }  
impl<T> FracGhost<T> {  
  spec fn id(self) -> Loc;  
  spec fn view(self) -> T;  
  spec fn frac(self) -> real;  
  proof fn split(...) -> ...;  
  ...  
}
```



Split the fractions between the thread and the shared state to coordinate updates
(make sure each thread's `view()` agrees with the shared states's `view()`)

Individual separation-logic-style permissions

```
tracked struct FracGhost<T> { ... }  
impl<T> FracGhost<T> {  
  spec fn id(self) -> Loc;  
  spec fn view(self) -> T;  
  spec fn frac(self) -> real;  
  proof fn split(...) -> ...;  
  ...  
}
```

thread a



thread_frac_a
frac() == 0.5

```
let (ptr, Tracked(atomicu32_perm)) = PAtomicU32::new(0);  
let tracked inv_frac_a = FracGhost::new(0);  
let tracked inv_frac_b = FracGhost::new(0);  
let tracked thread_frac_a = inv_frac_a.split();  
let tracked thread_frac_b = inv_frac_b.split();  
let tracked count_state = CountState { inv_frac_a, inv_frac_b, atomicu32_perm };  
let ghost inv_fn = |cs: CountState| {  
  ... cs.atomicu32_perm.value() == cs.inv_frac_a@ + cs.inv_frac_b@ };  
let tracked atomic_inv = AtomicInvariant::new(inv_fn, count_state, ...);
```

thread b



thread_frac_b
frac() == 0.5

shared AtomicU32
with AtomicInvariant

inv_frac_a
frac() == 0.5

inv_frac_b
frac() == 0.5

Individual separation-logic-style permissions

```
let (ptr, Tracked(atomicu32_perm)) = PAtomicU32::new(0);
let tracked inv_frac_a = FracGhost::new(0);
let tracked inv_frac_b = FracGhost::new(0);
let tracked thread_frac_a = inv_frac_a.split();
let tracked thread_frac_b = inv_frac_b.split();
let tracked count_state = CountState { inv_frac_a, inv_frac_b, atomicu32_perm };
let ghost inv_fn = |cs: CountState| {
    ... cs.atomicu32_perm.value() == cs.inv_frac_a@ + cs.inv_frac_b@ };
let tracked atomic_inv = AtomicInvariant::new(inv_fn, count_state, ...);

let thread_a = vstd::thread::spawn(|| {
    ... ptr, atomic_inv, thread_frac_a ...
});

let thread_b = vstd::thread::spawn(|| {
    ... ptr, atomic_inv, thread_frac_b ...
});

if let (Ok(join_a), Ok(join_b)) = (thread_a.join(), thread_b.join()) { ... }
```

Thread b performs fetch_add

thread a



thread_frac_a
frac() == 0.5
view() == 0

thread b



thread_frac_b
frac() == 0.5
view() == 0

shared AtomicU32
with AtomicInvariant

inv_frac_a
frac() == 0.5

inv_frac_b
frac() == 0.5
view() == 0

Thread b performs fetch_add

thread a



thread_frac_a
frac() == 0.5
view() == 0

thread b



shared AtomicU32
with AtomicInvariant

inv_frac_a
frac() == 0.5

inv_frac_b
frac() == 1.0
view() == 0

combine

Thread b performs fetch_add

thread a



thread_frac_a
frac() == 0.5
view() == 0

thread b



shared AtomicU32
with AtomicInvariant

inv_frac_a
frac() == 0.5

inv_frac_b
frac() == 1.0
view() == 1

update view()

Thread b performs fetch_add

thread a



thread_frac_a
frac() == 0.5
view() == 0

thread b



shared AtomicU32
with AtomicInvariant

inv_frac_a
frac() == 0.5

thread_frac_ibv_frac_b
frac() == 0.5 frac() == 0.5
view() == 1 view() == 1

split

Thread b performs fetch_add

thread a

thread b

shared AtomicU32
with AtomicInvariant



```
let thread_b = vstd::thread::spawn(|| { ...  
  open_atomic_invariant!(atomic_inv.borrow() => count_state => {  
    proof {  
      count_state.inv_frac_b.combine(thread_frac_b);  
    }  
    ptr.fetch_add(Tracked(&mut count_state.atomicu32_perm), 1);  
    proof {  
      count_state.inv_frac_b.update(1);  
      thread_frac_b = count_state.inv_frac_b.split();  
    }  
  });  
  Tracked(thread_frac_b)  
});
```

inv_frac_a
frac() == 0.5

inv_frac_b
frac() == 0.5
view() == 1

open_atomic_invariant can perform
1 exec operation + n proof operations

Parent thread joins thread a, thread b

thread a



thread_frac_a
frac() == 0.5
view() == 1

thread b



thread_frac_b
frac() == 0.5
view() == 1

parent thread



shared AtomicU32
with AtomicInvariant

inv_frac_a
frac() == 0.5
view() == ?

inv_frac_b
frac() == 0.5
view() == ?

Parent thread joins thread a, thread b

```
if let (Ok(join_a), Ok(join_b)) = (thread_a.join(), thread_b.join()) {  
  let Tracked(thread_frac_a) = join_a;  
  let Tracked(thread_frac_b) = join_b;  
  let n: u32;  
  open_atomic_invariant!(shared_atomic_inv.borrow() => count_state => {  
    n = ptr.load(Tracked(&mut count_state.atomicu32_perm));  
    proof {  
      count_state.inv_frac_a.agree(&thread_frac_a);  
      count_state.inv_frac_b.agree(&thread_frac_b);  
    }  
  });  
  assert(n == 2);  
}
```

shared AtomicU32
with AtomicInvariant

inv_frac_a
frac() == 0.5
view() == 1

inv_frac_b
frac() == 0.5
view() == 1



frac() == 0.5
view() == 1

frac() == 0.5
view() == 1

```
// Invariant tells us 1 + 1 == 2:  
let ghost inv_fn = |cs: CountState| { ...  
  cs.atomicu32_perm.value() ==  
    cs.inv_frac_a@ + cs.inv_frac_b@  
};
```

State machine for count-to-2

State type

```
counter: int  
inc_a: int  
inc_b: int
```

Globally, we have a protocol (a state machine) with an invariant
Locally:

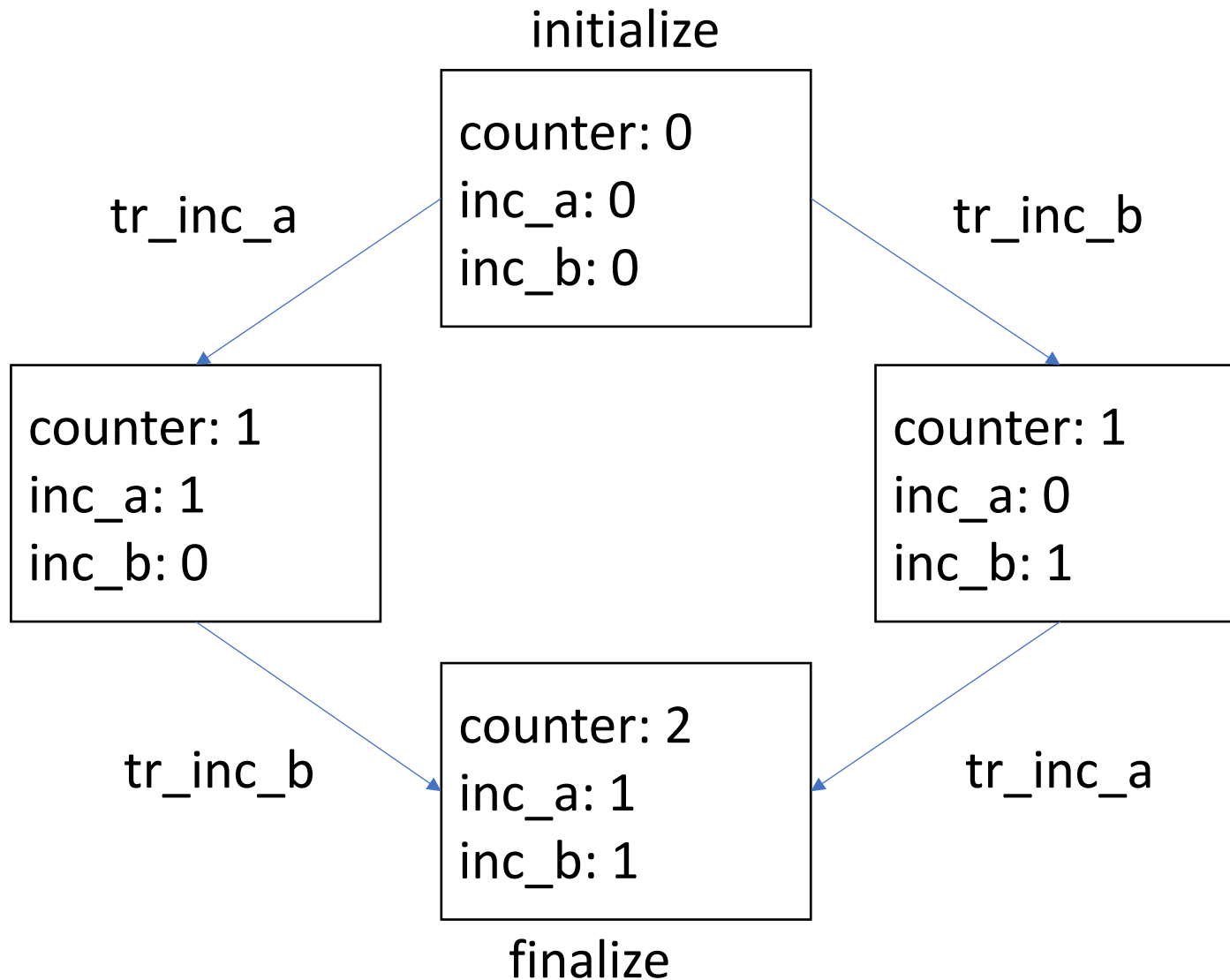
- thread_a is in charge of inc_a
- thread_b is in charge of inc_b

invariant:
 $\text{counter} = \text{inc_a} + \text{inc_b}$

Verus supports two different approaches to express global/local cooperation:

- Individual separation-logic-style permissions (a la Iris)
- **A dedicated macro-based state machine language**

State machine for count-to-2

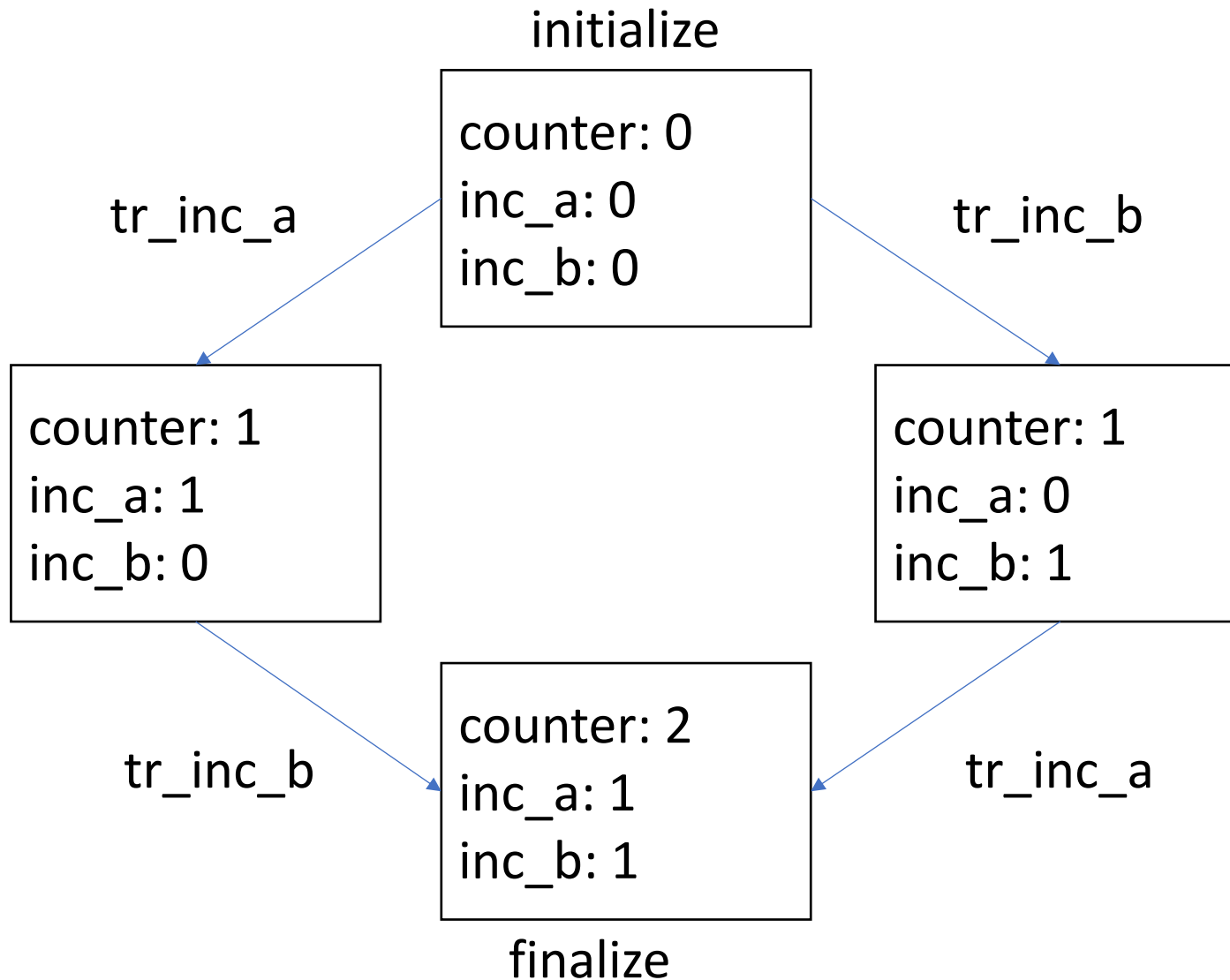


State type

```
counter: int  
inc_a: int  
inc_b: int
```

invariant:
 $\text{counter} = \text{inc_a} + \text{inc_b}$

State machine for count-to-2



```
// Slightly simplified to fit on slide
tokenized_state_machine!( Counter {
  fields {
    counter: int, inc_a: int, inc_b: int,
  }
  #[invariant] fn inv(&self) -> bool {
    &&& 0 <= self.inc_a < 2
    &&& 0 <= self.inc_b < 2
    &&& self.counter == self.inc_a + self.inc_b
  }
  init!{ initialize() {
    init counter = 0; init inc_a = 0; init inc_b = 0;
  }}
  transition!{ tr_inc_a() {
    require(pre.inc_a == 0);
    update counter = pre.counter + 1;
    update inc_a = 1;
  }}
  ...}
```

State machine for count-to-2

State machine generates
tracked permissions for threads

```
...
let tracked (
    Tracked(instance),
    Tracked(counter),
    Tracked(inc_a),
    Tracked(inc_b),
) = Counter::Instance::initialize();
...
let thread_a = vstd::thread::spawn(|| {
    ... inc_a ...
});
let thread_b = vstd::thread::spawn(|| {
    ... inc_b ...
});
...
```

```
// Slightly simplified to fit on slide
tokenized_state_machine!( Counter {
    fields {
        counter: int, inc_a: int, inc_b: int,
    }
    #[invariant] fn inv(&self) -> bool {
        &&& 0 <= self.inc_a < 2
        &&& 0 <= self.inc_b < 2
        &&& self.counter == self.inc_a + self.inc_b
    }
    init!{ initialize() {
        init counter = 0; init inc_a = 0; init inc_b = 0;
    }}
    transition!{ tr_inc_a() {
        require(pre.inc_a == 0);
        update counter = pre.counter + 1;
        update inc_a = 1;
    }}
    ...}
}
```

Any questions?